

YOOBEE COLLEGE OF CREATIVE INNOVATION

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Master of Business Informatics (Level 9)

MBI908: Industry-based Capstone Research Project

**Business Intelligence and Explainable Artificial
Intelligence Decision Support System for Customer
Retention in Retail SMEs**

**Integrating Customer Churn Prediction, Behavioural Segmentation, SHAP
Explainability, and Counterfactual Recommendations within a Unified Decision
Support System**

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Executive Summary

This capstone project presents the design, development, and evaluation of a Business Intelligence and Explainable Artificial Intelligence (XAI) Decision Support System to support customer retention decisions for retail small and medium-sized enterprises (SMEs). The research addresses the challenge of customer churn by integrating predictive analytics with explainable machine learning to improve the transparency and practical usefulness of retention recommendations.

The developed prototype follows a Design Science Research methodology and combines a Random Forest churn prediction model with customer behavioural segmentation based on an adapted RFM framework, where CashbackAmount is used as a proxy for monetary value due to dataset limitations. To enhance decision transparency, the system incorporates SHAP explainability to identify the factors influencing individual churn predictions and DiCE counterfactual explanations to generate actionable recommendations for reducing churn risk. These components are integrated into an interactive Streamlit dashboard that provides business users with customer profiles, churn risk predictions, behavioural insights, retention priorities, and personalised decision support.

The prototype was evaluated through both technical performance assessment and user evaluation involving business students, retail managers, a small business owner, and a data engineer. The findings indicate that combining predictive analytics with explainable AI improves the interpretability and usability of machine learning outputs, supporting more informed and transparent customer retention decisions. While the research acknowledges limitations related to the dataset, prototype scope, and evaluation sample, it demonstrates the practical value of integrating Business Intelligence and Explainable AI within a unified decision support system for retail SMEs. The study contributes a human-centred and explainable approach to AI-enabled decision support while identifying opportunities for future enhancement and real-world deployment.

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Chapter 1 – Introduction

1.1 Chapter Introduction

This chapter establishes the context and direction of the research. It outlines the background to customer retention analytics, defines the industry problem affecting retail SMEs, and presents the research aim, objectives, questions, scope, and report structure.

1.2 Background

Customer retention contributes to organisational stability by supporting repeat purchasing, customer lifetime value, and more efficient use of marketing resources (Berry & Linoff, 2004). Advances in Business Intelligence and machine learning have enabled organisations to analyse customer behaviour and identify customers at risk of churn before disengagement occurs. However, the practical value of predictive analytics depends on whether decision-makers can interpret model outputs and translate them into appropriate business actions. Explainable Artificial Intelligence addresses this requirement by providing insight into the factors influencing predictions and supporting more transparent human decision-making.

1.3 Industry Problem

Retail SMEs often lack the analytical expertise, financial capacity, and integrated information systems required to operationalise advanced customer churn models. Although predictive models can identify customers with elevated churn risk, prediction probabilities alone provide limited guidance regarding why a customer has been classified as at risk, which customers should be prioritised, or what retention action should be considered. This creates a gap between analytical prediction and managerial intervention. The research addresses this industry problem by developing an integrated decision support artefact that combines churn prediction, behavioural segmentation, explainability, prioritisation, and actionable recommendations.

1.4 Research Aim

The aim of this research is to design, develop, and evaluate a Business Intelligence and Explainable Artificial Intelligence (XAI)-enabled decision support system that assists retail SMEs in identifying customers at risk of churn and supports transparent, evidence-based retention decision-making.

1.5 Research Objectives

This research was guided by the following objectives:

1. To investigate the application of Business Intelligence, machine learning, and Explainable Artificial Intelligence (XAI) techniques for customer retention in retail SMEs.
2. To design and develop an integrated decision support system that combines customer churn prediction, RFM customer segmentation, SHAP explainability, and counterfactual recommendations.

3. To evaluate the predictive performance, explainability, and practical usefulness of the proposed system using technical performance measures and user-centred evaluation.
4. To examine the ethical, cultural, and governance considerations associated with the responsible adoption of AI-enabled decision support systems.

1.6 Research Questions

This research addresses the following research questions:

RQ1: How can SHAP-based Explainable Artificial Intelligence (XAI) support managerial understanding of customer churn predictions in retail SMEs?

RQ2: How can counterfactual explanation techniques support the generation of actionable customer retention recommendations within a customer churn prediction system?

RQ3: How can an Explainable AI-enabled decision support system support customer retention decision-making within retail SMEs?

1.7 Scope of the Study

This study is limited to the development and formative evaluation of a prototype using a publicly available e-commerce customer churn dataset. Evaluation covers predictive performance, model explainability, recommendation outputs, interface functionality, and user perceptions of usability and practical usefulness. The research does not include live organisational deployment, integration with CRM or enterprise data systems, measurement of actual retention outcomes, or longitudinal assessment of organisational adoption. Consequently, the findings demonstrate prototype feasibility and perceived decision-support value rather than proven improvement in customer retention or business performance.

1.8 Report Structure

This report comprises eight chapters. Chapter 1 introduces the research context, objectives, and scope. Chapter 2 reviews the relevant literature and identifies the research gap. Chapter 3 describes the research methodology. Chapter 4 presents the design and implementation of the proposed decision support system. Chapter 5 evaluates the developed artefact using technical performance measures and user evaluation. Chapter 6 critically reflects on the ethical and cultural considerations, together with the technical and practical limitations of the research. Chapter 7 discusses the research findings, contributions, and business implications. Finally, Chapter 8 concludes the research by summarising the key findings, contributions, limitations, and future research directions.

1.9 Chapter Summary

This chapter established the research problem, aim, objectives, questions, and scope. Chapter 2 critically examines the relevant literature and identifies the research gap underpinning the proposed decision support system.

Chapter 2 — Literature Review

2.1 Introduction

Customer retention has become an important research area within Business Intelligence, customer analytics, and Artificial Intelligence because of its influence on organisational performance and long-term competitiveness. As organisations increasingly rely on customer data, predictive analytics has improved the ability to identify customers at risk of churn. More recently, attention has shifted towards Explainable Artificial Intelligence (XAI), recognising that predictive performance alone is insufficient unless decision-makers can understand and act on model outputs. This chapter critically reviews the literature underpinning the research and identifies the gap addressed by the proposed decision support system.

2.2 Customer Retention and Customer Churn in Retail SMEs

Customer retention is recognised as a strategic priority because retaining existing customers generally requires fewer resources than acquiring new ones while contributing to customer lifetime value and long-term profitability. This challenge is particularly significant for retail SMEs, where limited financial and analytical resources increase the importance of identifying customers at risk of churn at an early stage (Berry & Linoff, 2004; Mamun, 2025).

Customer churn is influenced by multiple behavioural and transactional factors, including purchasing frequency, customer tenure, complaint history, and engagement patterns. Consequently, organisations increasingly apply predictive analytics to identify customers who are likely to discontinue their relationship before churn occurs. However, previous research indicates that predictive accuracy alone is insufficient to support effective retention strategies because managers also require understandable insights that support practical retention decisions. This limitation has motivated the integration of Business Intelligence, decision support systems, and Explainable Artificial Intelligence to improve both predictive capability and managerial decision-making.

2.3 Business Intelligence and AI-Driven Decision Support

Business Intelligence (BI) has evolved from descriptive reporting systems towards decision support platforms that incorporate predictive analytics and machine learning to assist organisational decision-making. Contemporary BI systems enable organisations to transform large volumes of transactional data into actionable insights that support strategic and operational decisions. However, many AI-enabled BI applications continue to prioritise predictive performance over transparency, limiting managerial understanding of how analytical outcomes are generated (Sharda et al., 2018).

Decision Support Systems (DSS) extend the capabilities of BI by combining analytical models with business knowledge to assist managers in evaluating alternative actions. Recent advances in Artificial Intelligence have enhanced DSS through improved predictive capability, yet the increasing complexity of machine learning models has also introduced challenges relating to explainability, accountability, and organisational trust. Consequently, researchers increasingly argue that effective AI-enabled decision support should balance predictive accuracy with transparency and interpretability to ensure that analytical outputs can be confidently incorporated into managerial decision-making (Sutton et al., 2020).

2.4 Explainable Artificial Intelligence for Decision Support

2.4.1 Explainable Artificial Intelligence

Machine learning models increasingly support organisational decision-making but often operate as "black boxes", making it difficult for users to understand how predictions are generated. This lack of transparency can limit the practical adoption of AI-assisted decision support (Adadi & Berrada, 2018).

Explainable Artificial Intelligence (XAI) addresses these challenges by providing interpretable explanations that enable users to understand, validate, and appropriately act upon predictive outputs without substantially compromising model performance (Arrieta et al., 2019). Consequently, explainability has become a key principle of responsible AI and is recognised within international governance frameworks such as the OECD AI Principles and the NIST Artificial Intelligence Risk Management Framework, both of which emphasise transparency, human oversight, and trustworthy AI (National Institute of Standards and Technology, 2023; Organisation for Economic Co-operation and Development, 2019). Among the available explainability techniques, SHAP and counterfactual explanations have emerged as complementary approaches for supporting transparent and actionable business decision-making.

2.4.2 SHAP Explanations

SHapley Additive exPlanations (SHAP) is one of the most widely adopted post-hoc explainability techniques for interpreting complex machine learning models, particularly when analysing structured tabular data. SHAP provides both global explanations that identify the overall importance of predictive features and local explanations that explain the factors influencing an individual prediction (Lundberg & Lee, 2017). This dual capability makes SHAP particularly valuable in customer churn prediction, where organisations require both strategic insight into overall churn drivers and customer-specific explanations to support operational decision-making.

Compared with other post-hoc explainability techniques, such as LIME (Local Interpretable Model-agnostic Explanations) and Partial Dependence Plots (PDPs), SHAP provides more consistent and theoretically grounded feature attribution based on Shapley values from cooperative game theory (Lundberg & Lee, 2017; Ribeiro et al., 2016). Consequently, SHAP has become one of the most widely adopted explainability techniques for structured business data because it improves transparency while preserving the predictive performance of complex machine learning models.

Despite these advantages, SHAP explains how features contribute to model predictions rather than how future outcomes may be changed. Furthermore, SHAP explanations represent associations learned by the predictive model and should not be interpreted as evidence of causal relationships (Molnar, 2024). Consequently, additional approaches are required to generate actionable recommendations for decision-makers, leading to increasing interest in counterfactual explanations.

2.4.3 Counterfactual Explanations

Counterfactual explanations complement feature attribution methods by identifying feasible changes that could alter a model's prediction rather than simply explaining why a prediction occurred (Wachter et al., 2017). Counterfactual explanations answer questions such as "What changes would be required for this customer to be classified as low churn risk instead of high churn risk?" This action-oriented perspective makes counterfactual explanations particularly valuable in business environments where managers must evaluate potential interventions rather than merely interpret predictive outputs.

In customer retention, counterfactual explanations can support decision-making by suggesting practical behavioural changes or engagement strategies that are associated with lower predicted churn risk. Counterfactual explanations therefore complement feature attribution by supporting intervention planning rather than model interpretation alone. Recent studies argue that combining feature attribution techniques with counterfactual reasoning provides a more comprehensive approach to explainability because managers require both an understanding of current predictions and guidance regarding possible future actions (Guidotti et al., 2018; Molnar, 2024).

Despite these advantages, counterfactual explanations remain dependent on the underlying predictive model and should be interpreted as decision-support guidance rather than guaranteed business outcomes. Recommended actions may also be constrained by organisational resources, customer behaviour, and operational feasibility. Therefore, counterfactual explanations should complement managerial judgement rather than replace it.

2.4.4 Synthesis of Explainability Techniques

SHAP and counterfactual explanations provide complementary forms of explainability. SHAP identifies the factors contributing to model predictions, whereas counterfactual explanations identify feasible changes associated with alternative outcomes. Their integration offers both explanatory and action-oriented decision support, providing the conceptual foundation for the proposed system.

2.5 Customer Behaviour Analysis for Retention Decision-Making

Understanding customer behaviour is essential for developing effective retention strategies because customers differ in their purchasing patterns, engagement levels, and long-term value. Behavioural segmentation enables organisations to classify customers into meaningful groups, allowing marketing activities and retention initiatives to be tailored according to customer characteristics rather than applying uniform interventions (Berry & Linoff, 2004).

Recency, Frequency, and Monetary (RFM) analysis is one of the most widely adopted behavioural segmentation techniques in customer relationship management because it evaluates how recently customers purchased, how frequently they transact, and the monetary value they contribute to the organisation. Previous research has shown that RFM segmentation complements predictive analytics by providing additional business context that supports customer prioritisation and resource allocation (Berry & Linoff, 2004; Hughes, 1994). Rather than replacing churn prediction, behavioural segmentation enables organisations to distinguish between customers with similar churn risk but different strategic value, thereby improving decision support and retention planning.

Accordingly, this research integrates RFM segmentation with customer churn prediction and Explainable Artificial Intelligence to provide behavioural context for predictive analytics, thereby supporting more targeted and evidence-based customer retention decision-making.

2.6 Design Science Research

Design Science Research (DSR) is widely adopted within Information Systems research when the objective is to develop and evaluate innovative artefacts that address identified organisational problems (Peffer et al., 2007). Unlike explanatory research approaches that primarily seek to understand existing phenomena, DSR emphasises the creation of practical solutions while generating knowledge through artefact design, implementation, and evaluation.

DSR has been extensively applied in Business Intelligence and decision support research because it supports the integration of technical development with organisational evaluation. The methodology is particularly appropriate for this study as the research aims not only to investigate customer churn prediction but also to design, implement, and evaluate an Explainable AI-enabled decision support system capable of supporting customer retention decisions. Its application within this research is described in Chapter 3.

2.7 Literature Synthesis and Research Gap

Table 2.1

Literature Synthesis and Research Gap Analysis

Research Area	Contributions	Limitations
Customer churn prediction	High predictive accuracy	Limited interpretability and decision support
Business Intelligence & DSS	Data-driven decision-making	Limited explainability and actionable guidance
Explainable AI (including SHAP and counterfactuals)	Improves transparency through feature attribution and actionable explanations	Existing studies often evaluate explainability techniques separately rather than integrating them into business decision support
Customer behaviour segmentation	Supports customer prioritisation	Does not predict churn or explain predictions
Design Science Research	Supports artefact development and evaluation	Limited application to integrated XAI-enabled customer retention systems

Note. Adapted from the literature reviewed in this chapter (Berry & Linoff, 2004; Lundberg & Lee, 2017; Mamun, 2025; Peffer et al., 2007; Wachter et al., 2017).

The literature reviewed demonstrates substantial advances in customer churn prediction, Business Intelligence, Explainable Artificial Intelligence, and customer behaviour analysis. As summarised in Table 2.1, these approaches have improved predictive capability, decision support, and model interpretability; however, important limitations remain regarding managerial actionability and the integration of complementary analytical techniques.

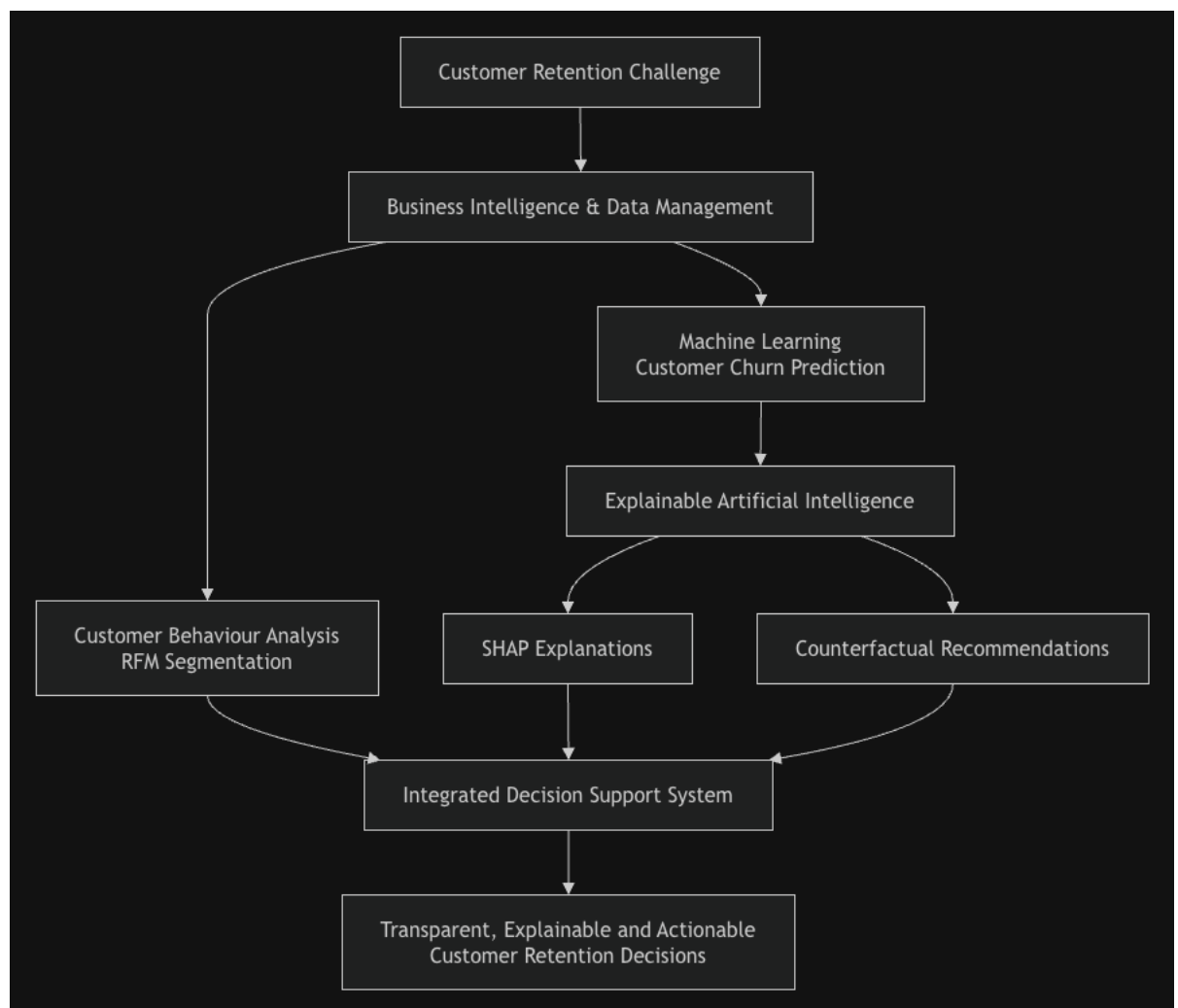
The review identifies a clear research opportunity to develop a unified decision support system that combines customer churn prediction, behavioural segmentation, SHAP explainability, counterfactual recommendations, and interactive visualisation within a single Business Intelligence environment for retail SMEs. This research addresses that gap through the design, development, and evaluation of an integrated Explainable AI-enabled decision support system.

2.8 Conceptual Framework for the Proposed Decision Support System

Based on the literature reviewed, this study adopts an integrated conceptual framework in which Business Intelligence, machine learning, Explainable Artificial Intelligence, and customer behaviour segmentation collectively support customer retention decision-making in retail SMEs. Figure 2.1 illustrates the relationships between these components and provides the theoretical foundation for the research methodology presented in Chapter 3.

Figure 2.1

Conceptual Framework for the Proposed Decision Support System



Note. Developed by the author based on the literature reviewed in Chapter 2.

Chapter 3 – Research Methodology

3.1 Introduction

This chapter presents the research methodology adopted to design, develop, and evaluate the proposed system for customer retention in retail SMEs. Guided by the Design Science Research (DSR) methodology, the research followed a structured implementation workflow encompassing dataset preparation, customer behaviour segmentation, predictive model development, explainability integration, recommendation generation, and prototype evaluation. The chapter explains the methodological rationale for each stage of the research, while detailed implementation and evaluation results are presented in Chapters 4 and 5.

3.2 Research Design

3.2.1 Design Science Research

This research adopted the Design Science Research (DSR) methodology because its primary objective was to design, develop, and evaluate an artefact that addresses an identified organisational problem rather than test an existing theory. DSR was appropriate because it enabled the research to combine technical implementation with practical evaluation, ensuring that the artefact was assessed according to both predictive performance and its usefulness in supporting managerial decision-making.

Design Science Research Methodology comprises six iterative activities: problem identification, objective definition, artefact design and development, demonstration, evaluation, and communication (Peffer et al., 2007).

Table 3.1

Mapping of the Research Project to Design Science Research Activities

DSR Activity (Peffer et al., 2007)	Implementation in This Research
Problem Identification	Identified the lack of transparent and actionable customer churn decision support for retail SMEs through the literature review and industry problem analysis.
Define Objectives	Established research objectives to develop an integrated Business Intelligence and Explainable AI decision support system with predictive, explainable, and recommendation capabilities.
Design and Development	Developed the artefact through data preparation, RFM segmentation, Predictive modelling, SHAP explainability, counterfactual recommendations, and a Streamlit dashboard.
Demonstration	Demonstrated the artefact using the customer churn dataset and interactive dashboard to generate customer predictions, explanations, and retention recommendations.
Evaluation	Evaluated the artefact using technical performance metrics, explainability outputs, and user evaluation involving eleven participants.

Communication	Communicated the research findings through this capstone report, presentation, and the developed software artefact.
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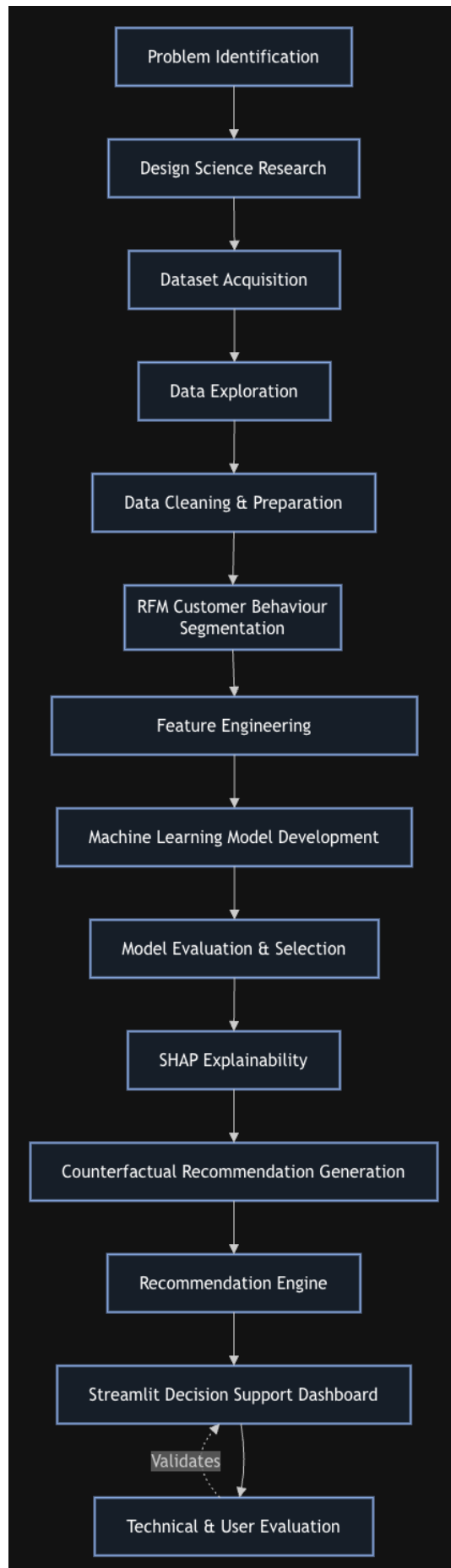
Note. Adapted from Peffers et al. (2007) and aligned with the implementation undertaken in this research. Rather than following a strictly linear process of DSR, several activities, particularly artefact development and evaluation, were performed iteratively as the prototype evolved through successive implementation and testing stages.

3.2.2 Research Process

The research followed the workflow illustrated in Figure 3.1, progressing from dataset exploration and preparation to predictive modelling, explainability integration, dashboard development, and artefact evaluation. Each stage informed the subsequent implementation process described in the following sections.

Figure 3.1

Research Workflow for the Design, Development, and Evaluation of the Proposed Decision Support System



Note. Developed by the author based on the Design Science Research methodology and the implemented research process.

3.3 Dataset Description

The research utilised a publicly available E-Commerce Customer Churn dataset obtained from Kaggle, containing transactional, demographic, and behavioural information relating to customers of an online retail organisation (Verma, n.d.). The dataset was selected because it contains a practically relevant combination of behavioural, service, demographic, and transactional attributes commonly used in customer churn analysis, including customer tenure, purchasing behaviour, complaint history, order frequency, cashback usage, satisfaction score, and other behavioural indicators. The target variable, Churn, identifies whether a customer discontinued their relationship with the organisation and therefore forms the basis for supervised classification. A complete description of all dataset variables used during preprocessing, segmentation, prediction, and explainability is provided in Appendix I.

The original dataset consisted of 5,630 customer records described by 20 predictor variables and one binary target variable (Churn). The variables included both numerical and categorical attributes representing customer demographics, purchasing behaviour, service interactions, and transaction history. Initial data profiling identified missing values in several variables, while duplicate records and class distribution were also examined during exploratory data analysis. These characteristics provided sufficient diversity to support customer behaviour segmentation, predictive modelling, and explainability analysis within the proposed decision support system.

Table 3.2

Summarises the Principal Characteristics of the Dataset prior to Preprocessing.

Dataset Characteristic	Description
Dataset source	Kaggle – E-Commerce Customer Churn Dataset (Verma, n.d.)
Number of records	5,630
Predictor variables	20
Target variable	Churn (Binary: Yes/No)
Variable types	Numerical and categorical
Data domains	Customer demographics, purchasing behaviour, service interactions, transaction history
Missing values	Present in selected numerical variables and treated during preprocessing
Duplicate records	Examined and removed during data cleaning
Class distribution	Assessed during exploratory data analysis before model development

Note. Summary based on the original dataset before preprocessing.

3.4 Data Preparation and Feature Engineering

Data preparation transformed the raw customer churn dataset into an analytical dataset suitable for predictive modelling and explainability. The process comprised exploratory analysis, data cleaning, behavioural segmentation, and machine learning preparation.

3.4.1 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted using Python libraries including Pandas, NumPy, Plotly and Matplotlib to assess dataset quality before model development. The analysis examined dataset structure, summary statistics, feature distributions, missing values, duplicate records, class imbalance, outliers, and relationships between customer attributes and churn using descriptive statistics and data visualisation techniques. Summary statistics and visualisations such as histograms, boxplots, correlation analysis, and class distribution plots were used to identify data quality issues and understand customer behaviour. These findings informed subsequent preprocessing decisions, including missing value treatment, feature transformation, behavioural segmentation, and machine learning preparation.

3.4.2 Data Cleaning and Feature Transformation

Data cleaning improved the consistency and reliability of the dataset prior to model development. Duplicate records were identified and removed, while missing numerical values were imputed using the median to minimise the influence of outliers. Categorical variables were transformed using one-hot encoding to create machine-learning-compatible features, and numerical variables were standardised where required for comparative model development. These preprocessing steps established a consistent analytical dataset that supported both predictive modelling and explainability.

3.4.3 Customer Behaviour Segmentation

Customer behaviour was analysed using the Recency, Frequency, and Monetary (RFM) framework to provide additional business context beyond churn prediction. RFM scores were calculated by assigning customer rankings for Recency, Frequency, and Monetary (CashbackAmount) variables before combining these scores to classify customers into behavioural segments representing different levels of customer value and engagement. As the dataset did not contain a direct monetary expenditure variable, the CashbackAmount associated with customer transactions was used as a proxy measure for the Monetary component of the RFM framework. Although cashback does not directly represent customer spending, it provides an indication of transaction value within the available dataset and enabled consistent behavioural segmentation. Behavioural segmentation provided additional context for prioritising customers with similar churn probabilities but different business value. A comprehensive description of the adapted RFM framework, behavioural customer segments, implemented retention-priority rules, and supporting decision logic is provided in Appendix C.

3.4.4 Machine Learning Preparation

Following feature engineering, the dataset was separated into predictor variables and the target variable before being partitioned into training and testing datasets using an 80:20 split. Numerical features were standardised using StandardScaler to ensure consistent feature scaling across the predictive models. To address the class imbalance observed in the training data, the

Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training dataset only, generating a balanced training set while leaving the testing dataset unchanged for unbiased model evaluation. The prepared dataset was subsequently used to train and compare multiple supervised machine learning algorithms under consistent experimental conditions.

3.5 Predictive Model Development

Following data preparation, multiple supervised machine learning algorithms were developed and evaluated to identify the most suitable predictive model for customer churn classification.

Model selection was guided by several complementary evaluation metrics, including accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). Because customer churn prediction represents an imbalanced classification problem, emphasis was placed on metrics that reflected the model's ability to correctly identify customers at risk of churn rather than relying exclusively on overall accuracy.

The selected predictive model subsequently formed the analytical foundation for the Explainable Artificial Intelligence components of the decision support system. Detailed comparative performance results and justification for the final model selection are presented in Chapter 5.

3.6 Explainability and Recommendation Strategy

The selected predictive model was integrated with SHAP and counterfactual explanations to provide both interpretability and actionable decision support. Their implementation is described in the following sections.

3.6.1 SHAP Explainability

SHAP was selected because it provides global explanations of overall model behaviour and local explanations for individual predictions, supporting transparent interpretation of customer churn predictions. Detailed results are presented in Chapter 5. The complete Explainable AI decision-support workflow, including the interaction between prediction, behavioural segmentation, SHAP explanations, counterfactual analysis, and managerial decision-making, is illustrated in Appendix D.

3.6.2 Counterfactual Recommendation Strategy

Counterfactual explanations were incorporated to extend explainability beyond interpretation by identifying practical changes associated with lower predicted churn risk. The implementation focused on actionable customer attributes while avoiding recommendations involving immutable characteristics. Additional guidance on interpreting dashboard outputs, counterfactual recommendations, and the role of human oversight is presented in Appendix D.

3.7 Evaluation Strategy

The artefact was evaluated using both technical and user-centred approaches, consistent with Design Science Research principles. Technical evaluation assessed predictive performance using standard classification metrics, while explainability was examined through SHAP analyses and counterfactual recommendations. Practical usefulness was evaluated through user feedback obtained following demonstration and hands-on interaction with the prototype. Combining these complementary evaluation approaches enabled assessment of both analytical

performance and business applicability. The principal methodological and technical design decisions that informed the development of the artefact, together with their implementation justifications, are summarised in Appendix B.

Chapter 4 System Design and Implementation

4.1 Introduction

This chapter presents the design and implementation of the proposed system for customer retention in retail SMEs. This chapter presents the design and implementation of the developed decision support artefact. It describes the system architecture, analytical pipeline, predictive engine, explainability components, recommendation engine, and Streamlit interface that together deliver customer retention decision support.

4.2 Overall System Architecture

Figure 4.1

Overall System Architecture of the Explainable AI-Enabled Business Intelligence Decision Support System



Note. Developed by the author based on the implemented system architecture.

Figure 4.1 illustrates the overall architecture of the developed decision support system. The artefact adopts a modular architecture that separates data preparation, predictive analytics, explainability, recommendation generation, and user interaction into interconnected

components. This design improves maintainability, reproducibility, and future extensibility by allowing each component to operate independently while contributing to a unified analytical workflow. A consolidated technical summary of the developed research artefact, including the implementation environment, analytical components, deployment platform, and supporting technologies, is provided in Appendix A.

The implementation begins with the customer churn dataset, which undergoes data preparation and customer behaviour segmentation before being processed by the predictive modelling component. The validated Random Forest model generates churn probability predictions that are subsequently enriched through an Explainable Artificial Intelligence layer comprising SHAP explanations and counterfactual recommendations. These analytical outputs are consolidated within the Recommendation Engine, which combines churn risk, behavioural segmentation, and business rules to prioritise customers and generate manager-oriented retention guidance.

The final analytical outputs are delivered through a Streamlit-based decision support interface that integrates customer profiles, retention priorities, explainability, and actionable recommendations within a single environment.

4.3 Business Intelligence Data Pipeline

The Business Intelligence data pipeline transforms raw customer data into validated analytical datasets that support the downstream components of the artefact. Following preprocessing and behavioural segmentation, the prepared data are supplied to the predictive model, after which prediction outputs are enriched through explainability and recommendation modules before being presented within the Streamlit interface. This pipeline ensures a consistent flow of information across all analytical components.

4.4 Predictive Modelling Implementation

4.4.1 Predictive Engine Implementation

The predictive engine was implemented using the processed datasets generated during the data preparation phase and formed the analytical core of the decision support system. To ensure a robust model selection process, five supervised machine learning algorithms were implemented using a common training and testing dataset: Logistic Regression, Random Forest, XGBoost, CatBoost, and a Soft Voting Ensemble. Applying a consistent analytical dataset across all candidate models ensured that subsequent performance differences reflected algorithmic capability rather than inconsistencies in data preparation.

Each model generated both binary churn classifications and prediction probability estimates. While binary predictions supported customer classification, probability outputs were retained because they were required by downstream components, including the Explainability Engine, Recommendation Engine, and Counterfactual Recommendation Module. This common prediction interface simplified integration between analytical modules and ensured that all components operated on consistent predictive outputs.

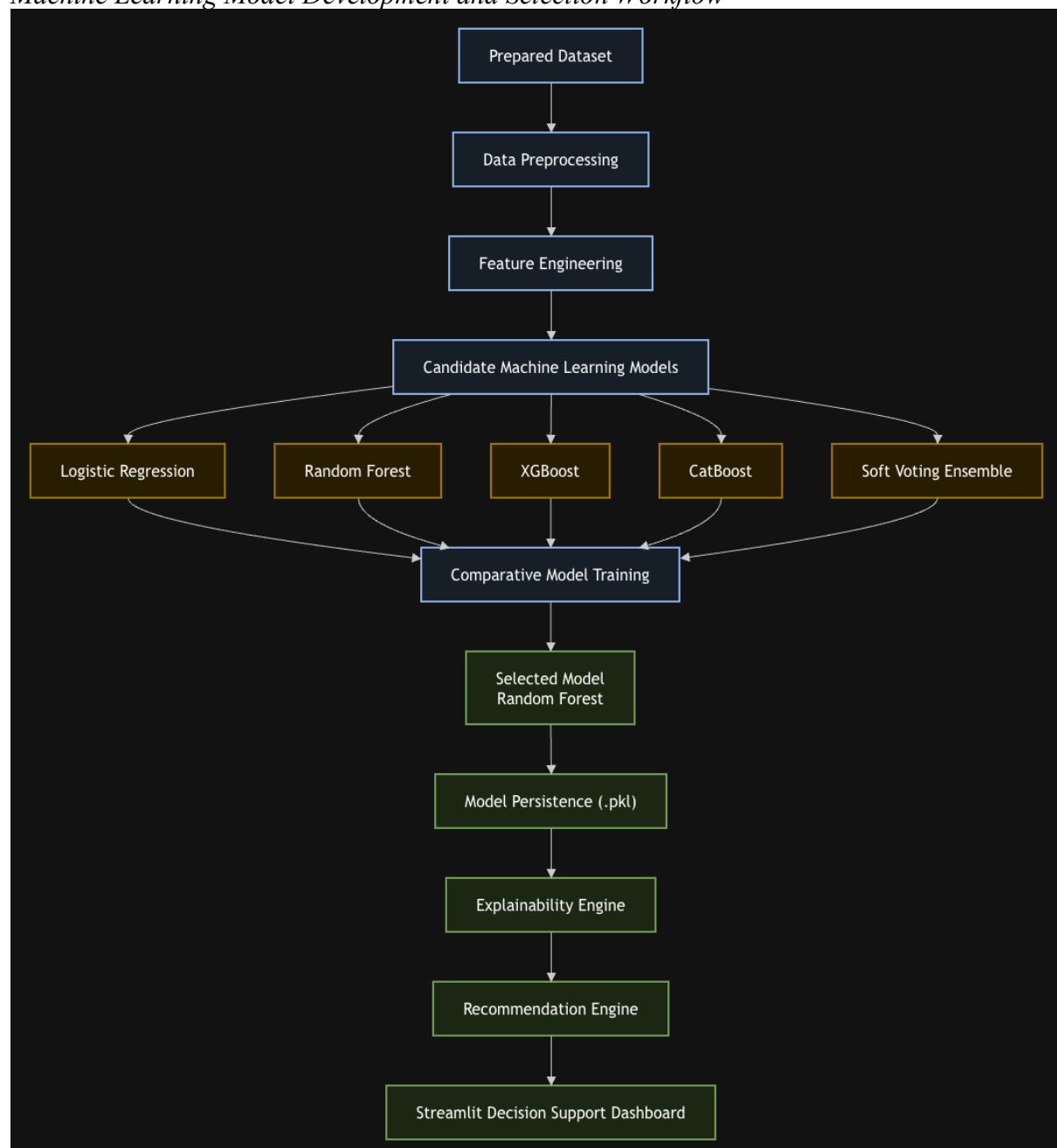
4.4.2 Model Persistence and Integration

Following comparative evaluation, the selected Random Forest model was serialised using the Joblib library and reused throughout the remaining implementation stages. Persisting the validated model separated model development from downstream analytical components, allowing the Explainability Engine, Recommendation Engine, Counterfactual Recommendation Module, and Streamlit dashboard to load the same predictive artefact without repeated training.

This architectural decision improved reproducibility, reduced computational overhead, and ensured consistency across the decision support system. Maintaining a single persisted predictive model also simplified future maintenance by enabling subsequent analytical components to operate independently while preserving identical predictive behaviour throughout the implemented artefact.

Figure 4.2

Machine Learning Model Development and Selection Workflow



4.5 Explainability Engine

The Explainability Engine was integrated directly with the validated Random Forest predictive model to improve the transparency of customer churn predictions. Rather than operating as an independent analytical module, explainability formed part of the end-to-end decision support workflow, ensuring that every explanation corresponded to the same predictive artefact used throughout the system. This design maintained consistency between prediction, interpretation, and recommendation generation while avoiding the need to retrain or duplicate analytical models.

SHAP TreeExplainer was employed to generate both organisation-level and customer-level explanations. Global explanations identified the features that most strongly influenced overall model behaviour, while local explanations supported interpretation of individual customer predictions. Integrating these complementary perspectives enabled managers to understand both general churn patterns and customer-specific risk factors within a single analytical framework.

To improve managerial relevance, the explanation and recommendation modules were configured to present only business-actionable variables. Although the predictive model utilised all selected features during classification, non-actionable demographic attributes, such as gender and marital status, were excluded from the SHAP explanations and counterfactual recommendations presented within the dashboard. This design enables managers to focus on customer characteristics that can realistically be influenced through retention strategies while preserving the predictive performance of the underlying model.

To improve usability, technical SHAP outputs were translated into concise business-readable explanations before being incorporated into the Recommendation Engine and Streamlit interface. This interpretation layer reduced the technical complexity of Explainable Artificial Intelligence and enabled analytical results to be understood without specialist machine learning knowledge.

4.6 Decision Support Engine

4.6.1 Recommendation Engine

The Decision Recommendation Engine integrates predictive outputs, customer behaviour segmentation, explainability results, and predefined business rules to generate prioritised customer retention recommendations. Rather than presenting individual analytical results independently, the engine consolidates multiple sources of evidence into a unified decision support process that assists managers in identifying customers requiring immediate retention intervention.

Customer churn probability, RFM segmentation, SHAP explanations, and business-defined priority rules are combined to classify customers according to both predicted churn risk and business value. This integration enables the system to distinguish customers with similar churn probabilities but different strategic importance, allowing retention resources to be directed towards customers likely to deliver the greatest business value.

Table 4.1*Business Rules Implemented in the Recommendation Engine*

Rule Component	Implemented Business Rule
Churn Risk Classification	High Risk (≥ 0.70), Medium Risk (0.40–0.69), Low Risk (< 0.40) based on predicted churn probability.
Customer Value Weighting	Additional priority score assigned according to RFM customer segment (e.g., High-Value Customers Losing Interest = +25; Previously Valuable Customers = +20; Frequent Customers Losing Interest = +15).
Priority Determination	High-risk customers receive Immediate Action. Medium-risk customers are classified as Immediate Action or Follow Up Soon depending on customer segment. Low-risk customers are classified as Monitor.
Manager Recommendation	Business-readable retention actions are generated using customer segment, churn risk, SHAP explanations, and counterfactual recommendations to support appropriate intervention strategies.

Note. Business rules implemented by the author based on the Recommendation Engine developed for this research.

The thresholds and segment weights were defined heuristically for prototype demonstration, informed by churn probability and relative RFM segment importance rather than validated intervention outcomes. They require organisational calibration using cost, customer lifetime value, campaign capacity, and observed retention outcomes before operational deployment. The complete operational priority classification rules and supporting pseudocode implemented within the dashboard are presented in Appendix C.

4.6.2 Counterfactual Recommendation Integration

Counterfactual explanations extend the Decision Recommendation Engine by identifying feasible behavioural changes associated with lower predicted churn risk. Rather than functioning as independent recommendations, counterfactual outputs are integrated with SHAP explanations and customer segmentation results to provide managers with both explanatory and action-oriented insights.

The implementation applies predefined feasibility constraints to ensure that generated recommendations focus on customer behaviours that can reasonably be influenced through retention strategies, such as purchasing frequency, engagement, or transaction recency. Combining counterfactual reasoning with predictive analytics enables the decision support system to move beyond risk identification towards supporting evidence-based intervention planning.

4.7 Integrated Decision Support Interface

The Streamlit interface provides the presentation layer of the developed artefact by consolidating prediction, behavioural segmentation, explainability, and recommendations within a single interactive environment for managerial decision support.

Customer profiles integrate demographic information, behavioural segmentation, churn predictions, SHAP explanations, retention priorities, and managerial recommendations within a single workspace.

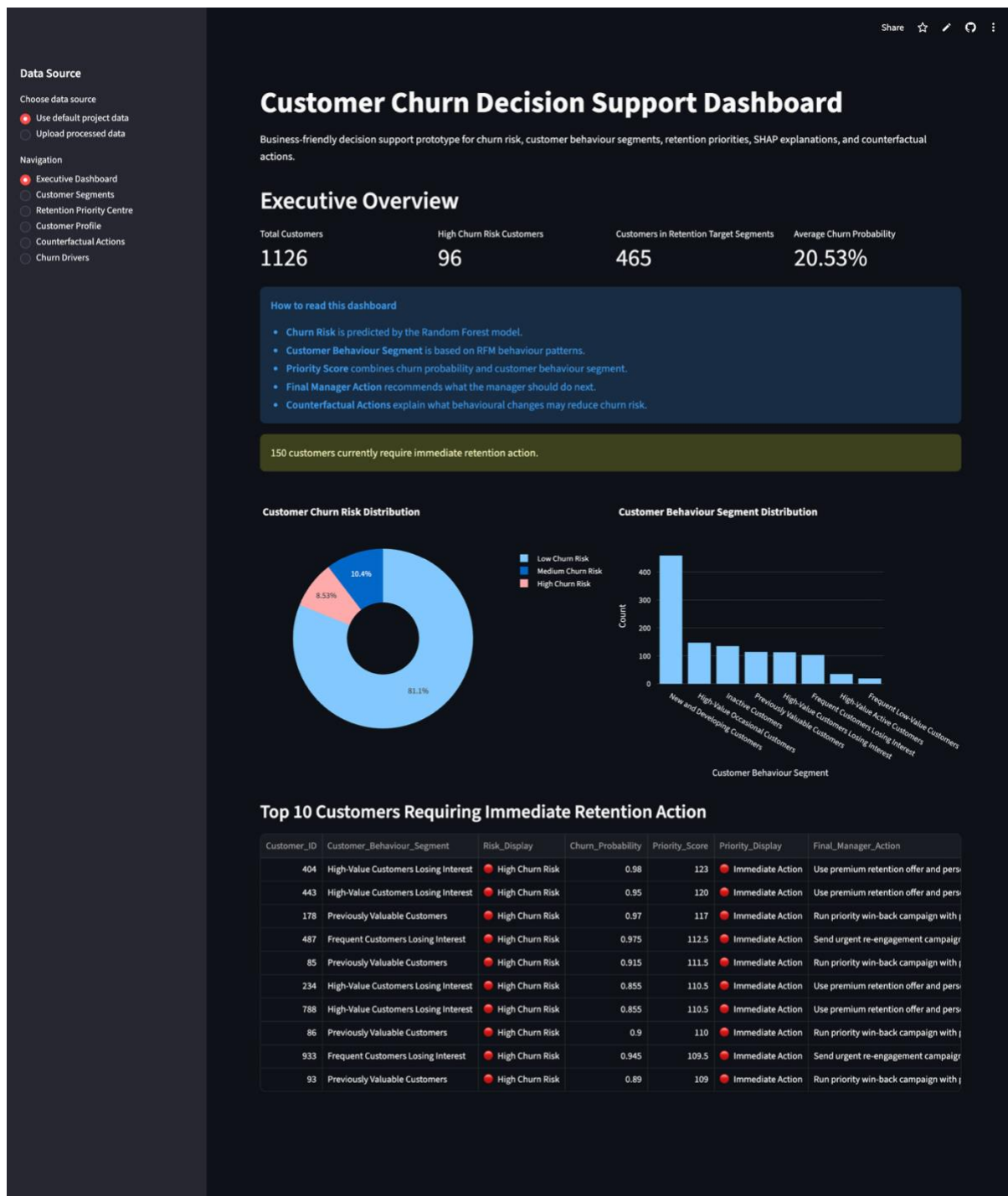
The interface was organised around a series of decision-support functions that address different managerial information requirements. Organisation-level monitoring is supported through executive summaries and customer segmentation views, while operational decision-making is facilitated through retention prioritisation, customer-level analysis, explanation visualisation, and actionable intervention planning. This functional organisation reflects the progression of customer retention decision-making, allowing managers to move naturally from strategic monitoring to detailed customer investigation and intervention planning.

Analytical outputs are presented in business-oriented language to improve interpretability for non-technical users.

4.7.1 Dashboard Overview

The Streamlit dashboard provides the primary interaction point between business users and the analytical components of the decision support system. Rather than presenting isolated analytical outputs, the interface integrates customer segmentation, churn prediction, explainability, and recommendation generation within a unified environment designed to support customer retention decision-making.

Figure 4.3
Main Dashboard Interface of the Developed Decision Support System



Note. Screenshot captured by the author from the implemented Streamlit decision support system.

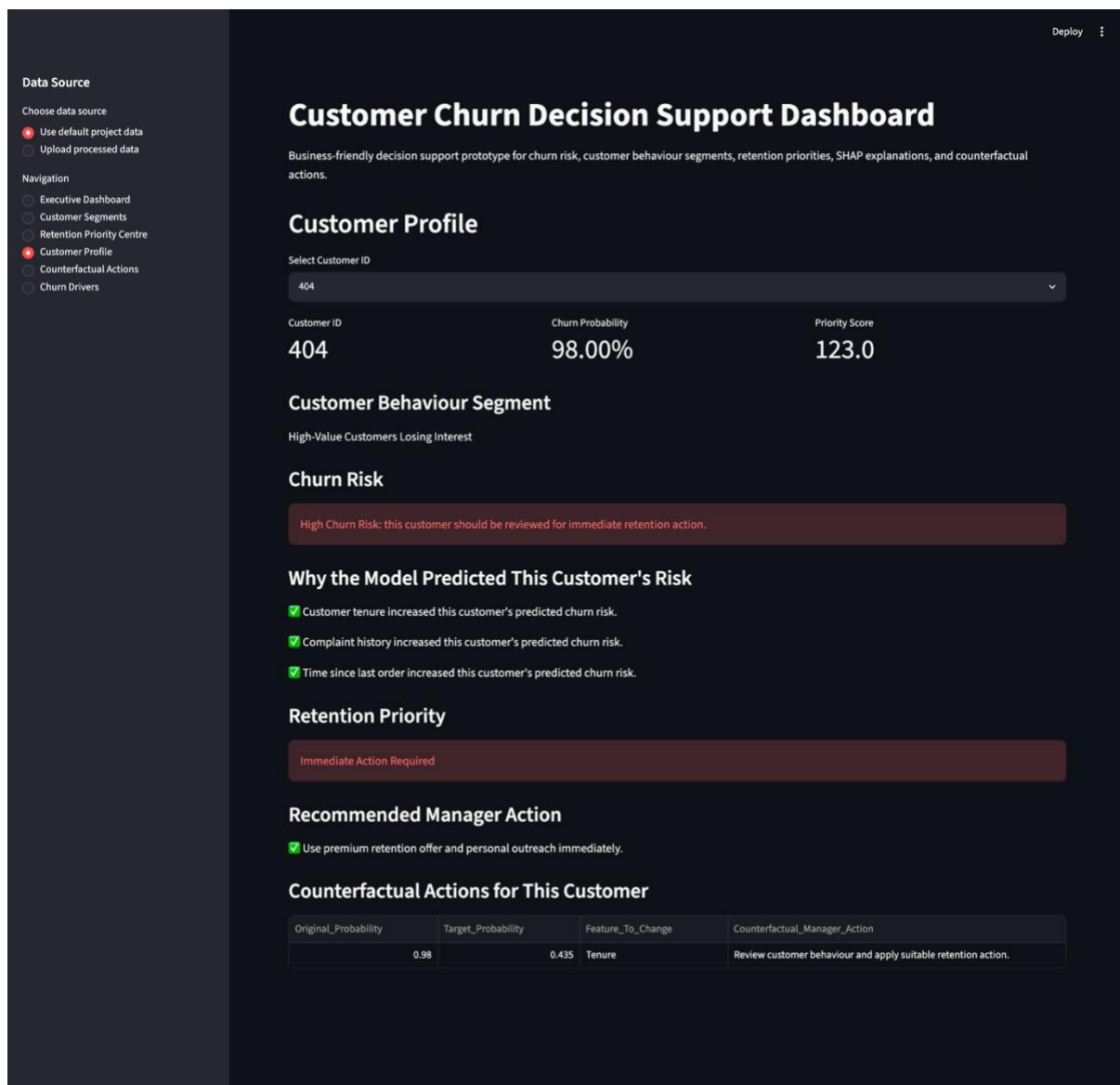
The dashboard organises analytical outputs into business-oriented views, enabling managers to review customer risk and access retention recommendations without interacting directly with the underlying machine learning models.

4.7.2 Customer Profile and Decision Support

The Customer Profile page consolidates customer-specific analytical information into a single decision support view. For each selected customer, the interface presents demographic information, behavioural segmentation, predicted churn risk, SHAP explanations, and prioritised retention recommendations. Integrating these complementary outputs enables managers to understand both the predicted outcome and the factors contributing to that prediction while identifying appropriate intervention strategies.

Figure 4.4

Individual Customer Profile and Decision Support Interface



Note. Screenshot captured by the author from the implemented Streamlit decision support system.

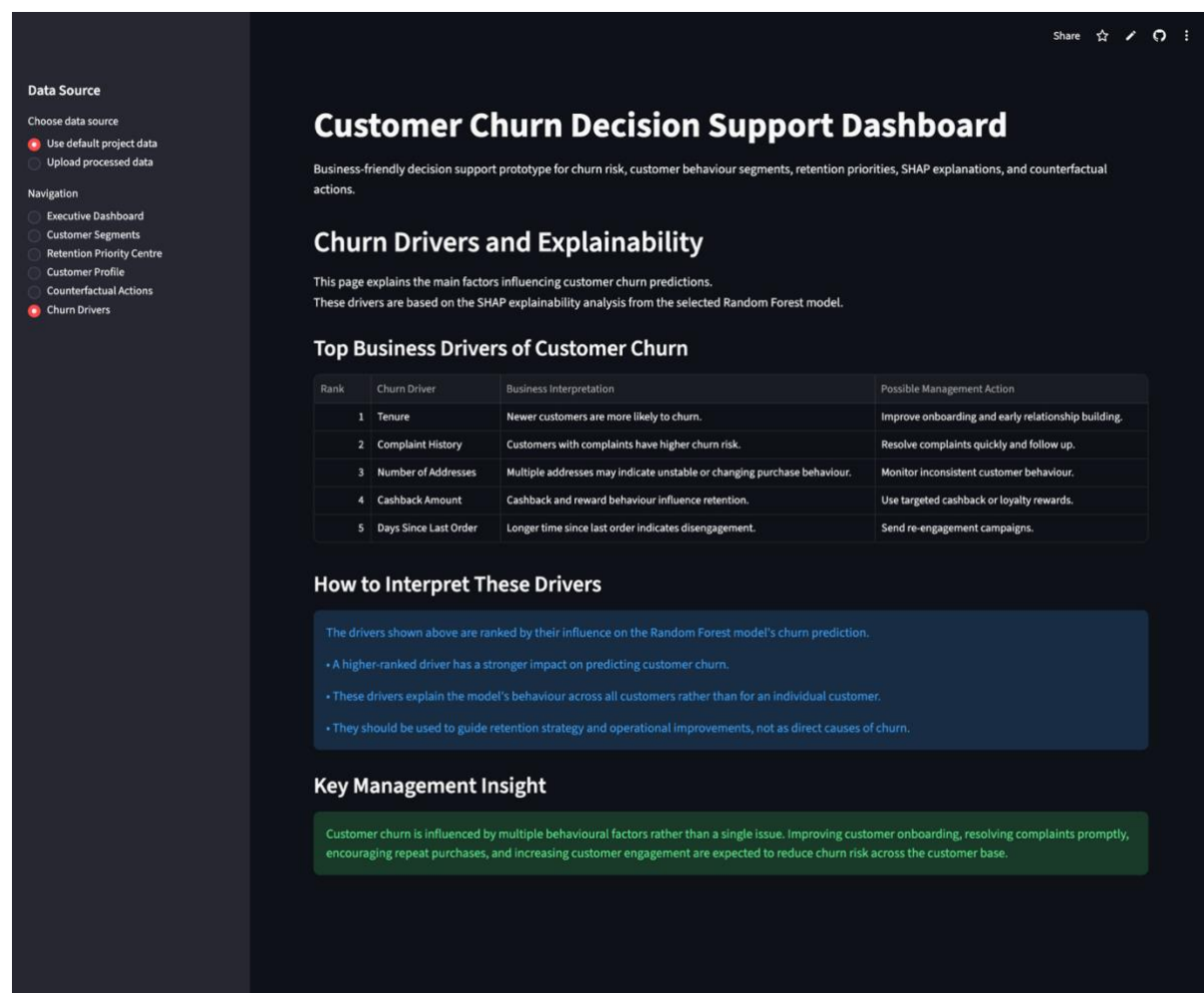
4.7.3 Explainability and Recommendation Interface

The explainability and recommendation interface integrates SHAP explanations with counterfactual recommendations to provide transparent and actionable decision support. Global and customer-level explanations help managers understand the principal drivers of predicted churn, while counterfactual recommendations identify feasible behavioural changes associated with lower churn risk. These outputs are presented alongside customer information and retention priorities to support evidence-based intervention planning.

Representative examples of the explainability and recommendation components are presented in Figures 4.5 and 4.6. Their effectiveness is evaluated in Chapter 5 – Results and Artefact Evaluation.

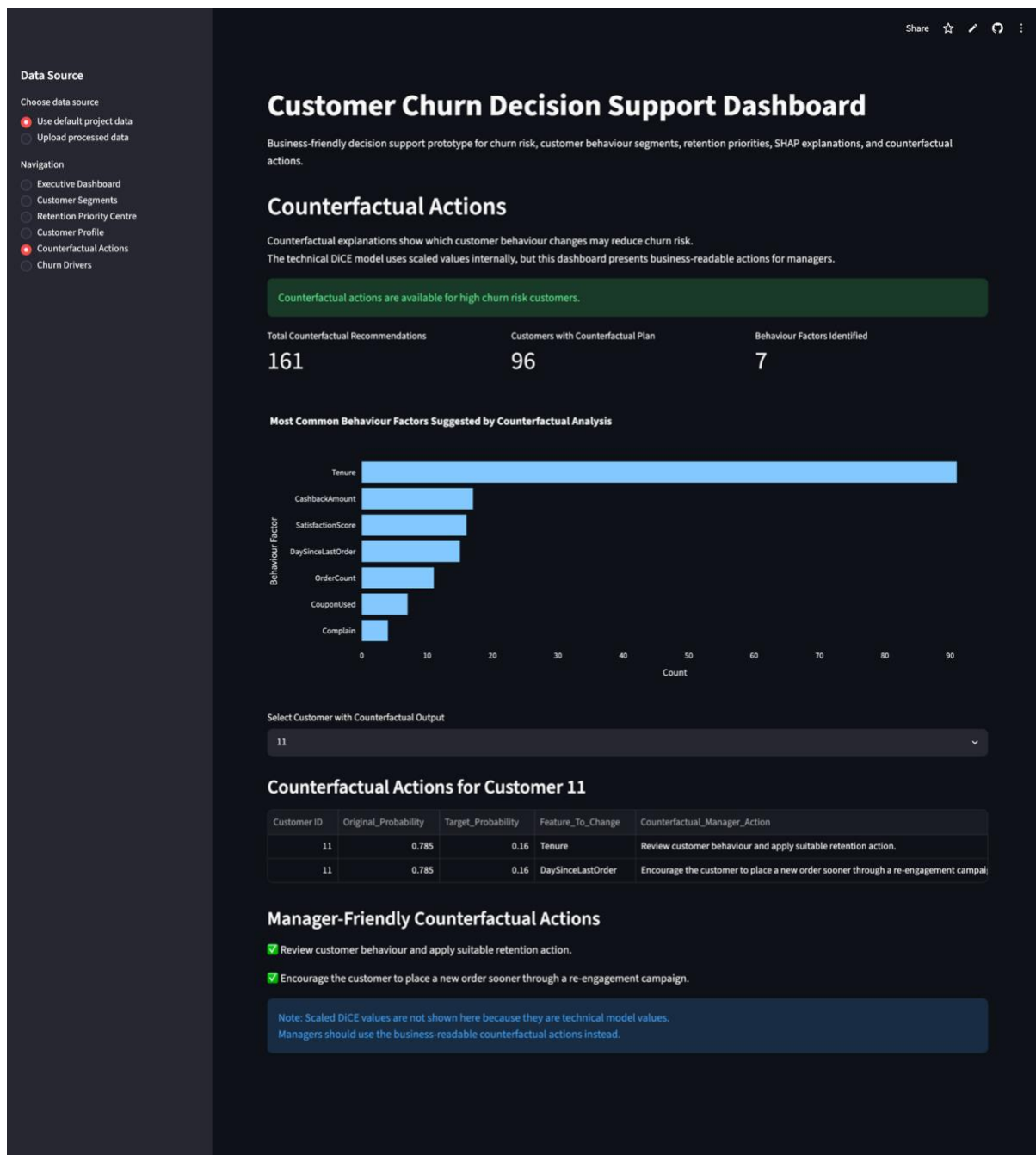
Figure 4.5

SHAP Explainability Interface for Customer-Level Prediction



Note. Screenshot captured by the author from the implemented Streamlit decision support system.

Figure 4.6
Counterfactual Recommendation Interface



Note. Screenshot captured by the author from the implemented Streamlit decision support system.

Additional screenshots illustrating implemented dashboard interfaces that were not included in the main discussion are provided in Appendix H.

4.8 Chapter Summary

This chapter described the overall system architecture, Business Intelligence data pipeline, predictive engine, Explainability Engine, Decision Support Engine, and Streamlit-based interface, demonstrating how these components were integrated into a unified analytical artefact. The following chapter evaluates the technical performance, explainability, and practical usefulness of the implemented system.

Chapter 5 – Results and Artefact Evaluation

5.1 Introduction

This chapter evaluates the developed Explainable AI-enabled Business Intelligence decision support system against the research objectives established in Chapter 1. Following the Design Science Research methodology, the evaluation considers both technical performance and practical usefulness. The assessment includes comparative machine learning results, predictive model performance, SHAP explainability, counterfactual recommendations, dashboard functionality, and user evaluation.

5.2 Evaluation of Predictive Capability

5.2.1 Comparative Model Performance

The predictive capability of the proposed system was evaluated by comparing five supervised machine learning algorithms using identical training and testing datasets, preprocessing procedures, and evaluation metrics. To reduce bias associated with the imbalanced class distribution, all candidate machine learning models were trained using the SMOTE-balanced training dataset, while evaluation was performed on the original unseen testing dataset. This ensured that performance differences reflected algorithm characteristics rather than inconsistencies in data preparation.

Table 5.1

Comparative Performance of Candidate Machine Learning Models

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	0.9618	0.9200	0.8474	0.8822	0.9935
XGBoost	0.9680	0.9375	0.8684	0.9016	0.9932
Soft Voting Ensemble	0.9627	0.8895	0.8895	0.8895	0.9892
CatBoost	0.9494	0.8757	0.8158	0.8447	0.9872

Logistic Regression	0.8028	0.4545	0.8421	0.5904	0.8877
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Although XGBoost achieved the highest accuracy (96.80%) and F1-score (0.9016), Random Forest produced a comparable accuracy (96.18%), the highest ROC-AUC (0.9935), and support efficient tree-based SHAP explanations, and offered a comparatively straightforward and reproducible integration path for the prototype’s explainability and downstream decision-support components.

5.2.2 Evaluation of the Selected Predictive Model

The selected Random Forest model was further evaluated using the independent testing dataset to confirm its suitability for integration with the explainability and recommendation modules.

Table 5.2

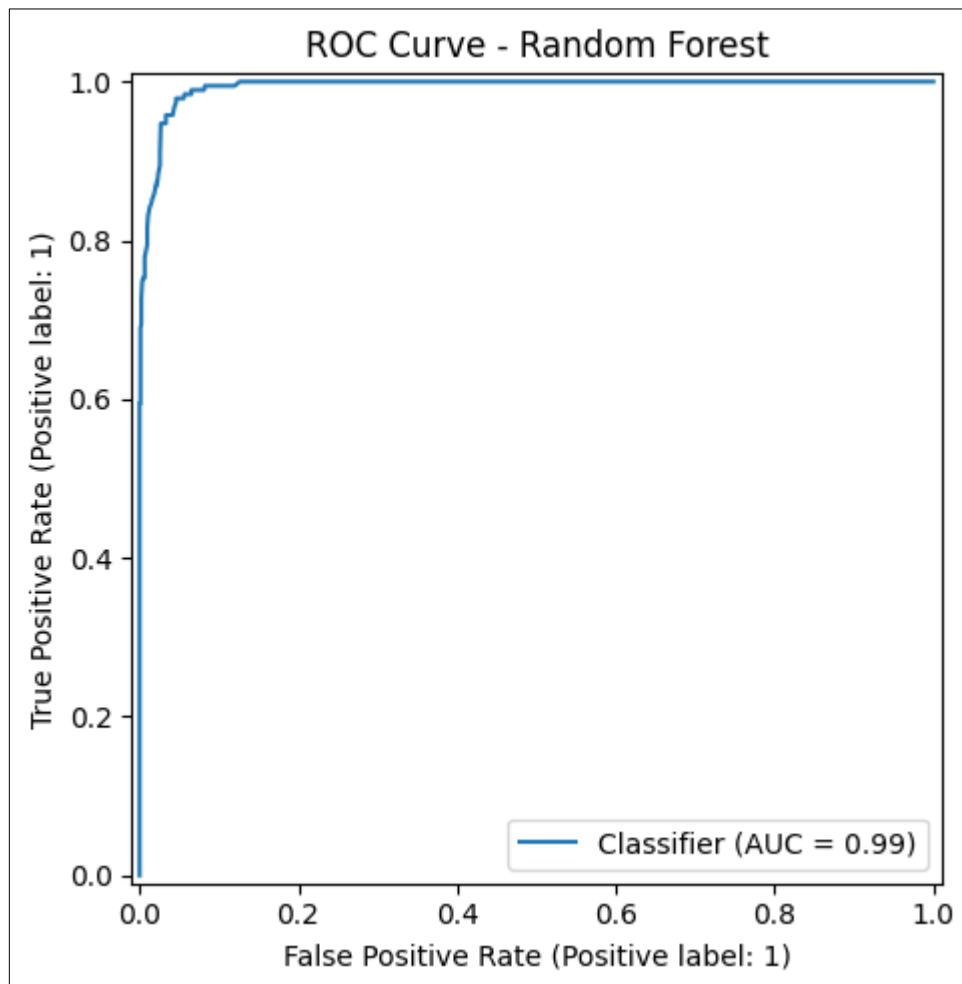
Performance Evaluation of the Selected Random Forest Model

Metric	Value
Accuracy	96.18%
Precision	92.00%
Recall	84.74%
F1-score	88.22%
ROC-AUC	0.994

The model achieved an accuracy of 96.18%, precision of 92.00%, recall of 84.74%, F1-score of 88.22%, and a ROC-AUC of 0.994, indicating excellent predictive performance.

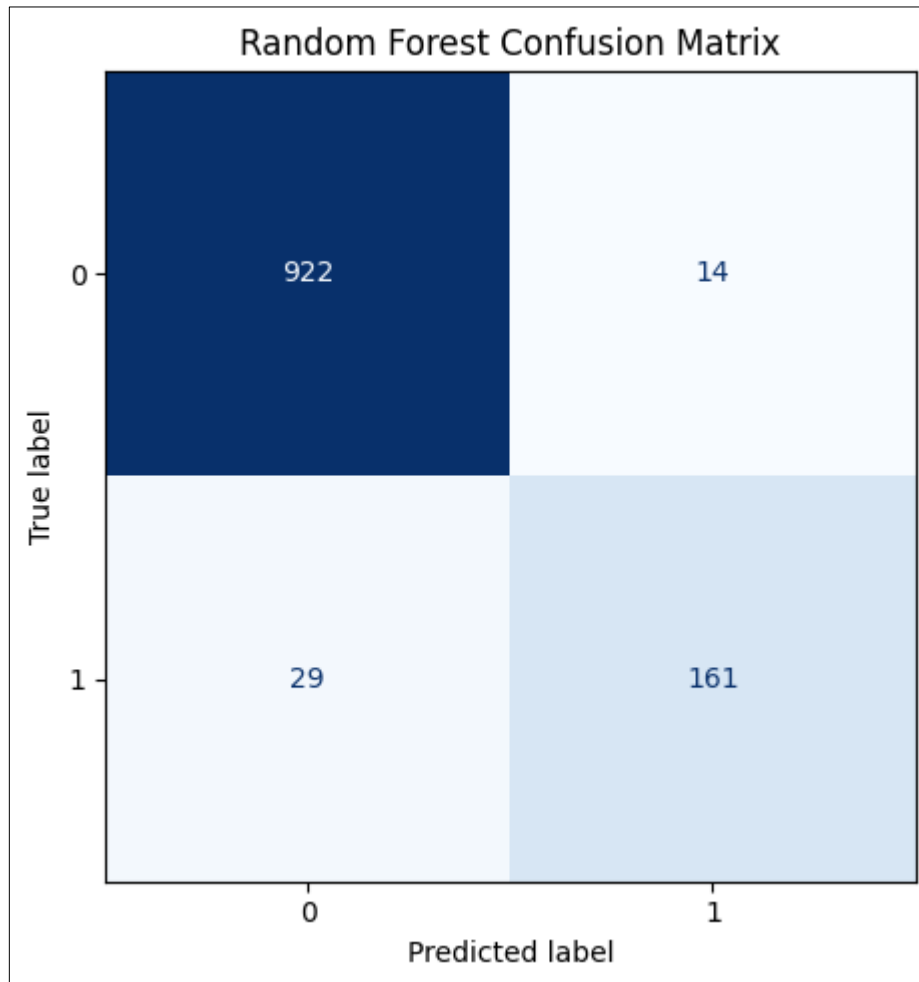
Figure 5.1

ROC Curve of the Selected Random Forest Model



The ROC curve demonstrates strong discriminative capability, with performance remaining close to the upper-left corner of the graph. This indicates that the model effectively distinguishes between customers likely to churn and those likely to remain.

Figure 5.2
Confusion Matrix for the Selected Random Forest Model



The confusion matrix shows that the model correctly classified 922 retained customers and 161 churned customers, with only 43 misclassifications (14 false positives and 29 false negatives). The classification report indicates a weighted F1-score of 0.96, reflecting excellent overall performance across both classes. For the churn class, the model achieved an F1-score of 0.88, demonstrating a strong balance between precision (0.92) and recall (0.85) in identifying customers at risk of churn.

These results indicate that the model provides reliable predictive performance suitable for subsequent explainability and recommendation analyses.

5.3 Evaluation of Explainability Engine

Following validation of the predictive model, the Explainability Engine was evaluated through global feature importance, customer-level explanations, and business-readable outputs.

5.3.1 Global Explainability

Global explainability was assessed using SHAP feature importance and SHAP summary visualisations to identify the variables that most strongly influenced customer churn predictions.

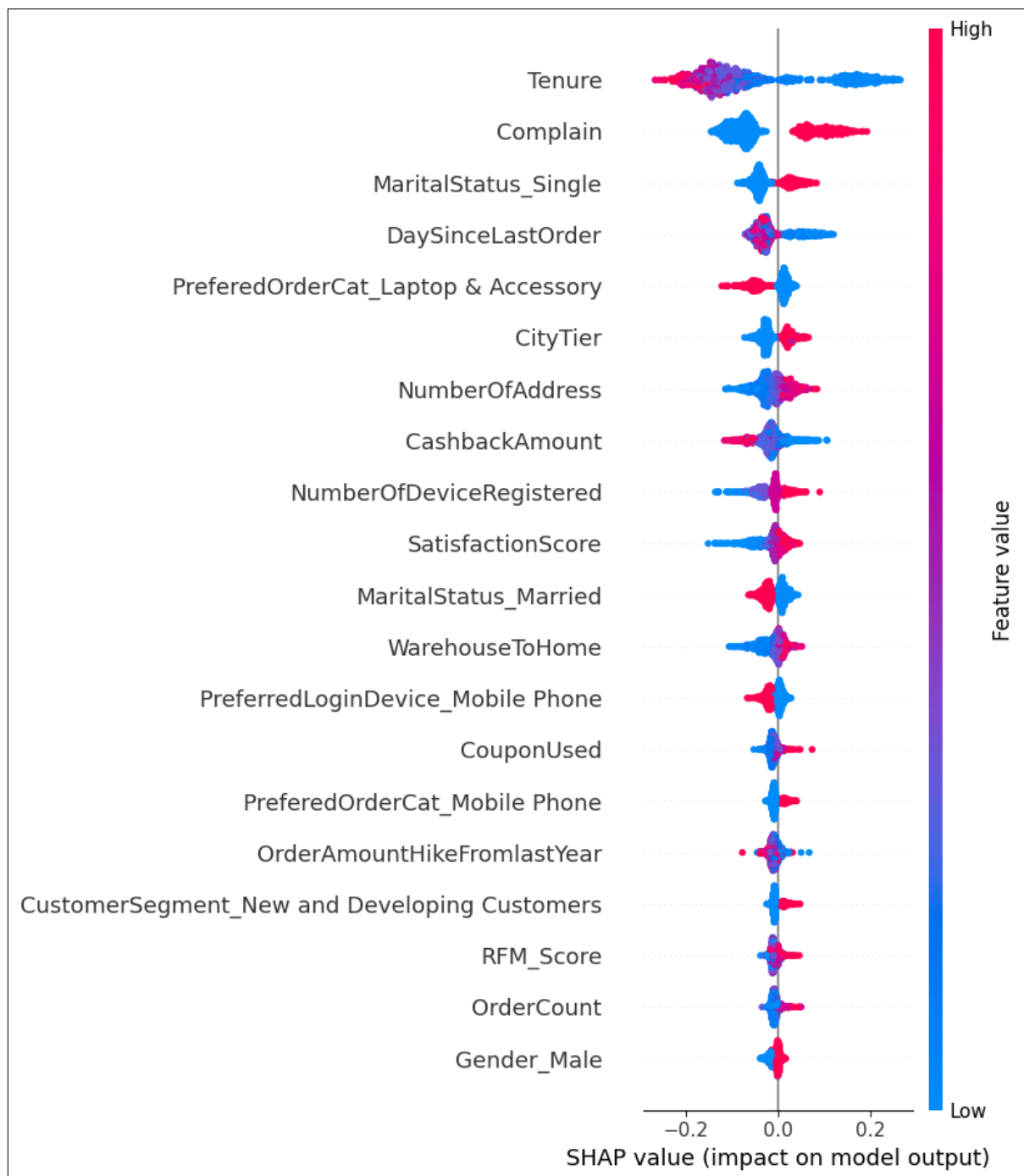
Figure 5.3.

Global SHAP feature importance for the selected Random Forest model.



Figure 5.4

SHAP summary plot illustrating the influence and direction of feature contributions across the testing dataset



The global SHAP analysis identified Tenure, Complain, MaritalStatus (Single), DaySinceLastOrder, and PreferredOrderCat (Laptop & Accessory) as the most influential predictors of customer churn. The SHAP summary plot further demonstrated that the influence of these variables varied across customers, indicating that churn behaviour resulted from multiple interacting factors rather than a single dominant predictor.

These findings provide valuable business insight by highlighting the behavioural characteristics most strongly associated with customer churn.

5.3.2 Customer-Level Explainability

While global explainability supports strategic understanding, customer retention decisions are ultimately made at the individual customer level. The Explainability Engine was therefore evaluated to determine whether it could generate meaningful explanations for individual predictions.

Figure 5.5

Local SHAP explanation for an individual customer prediction

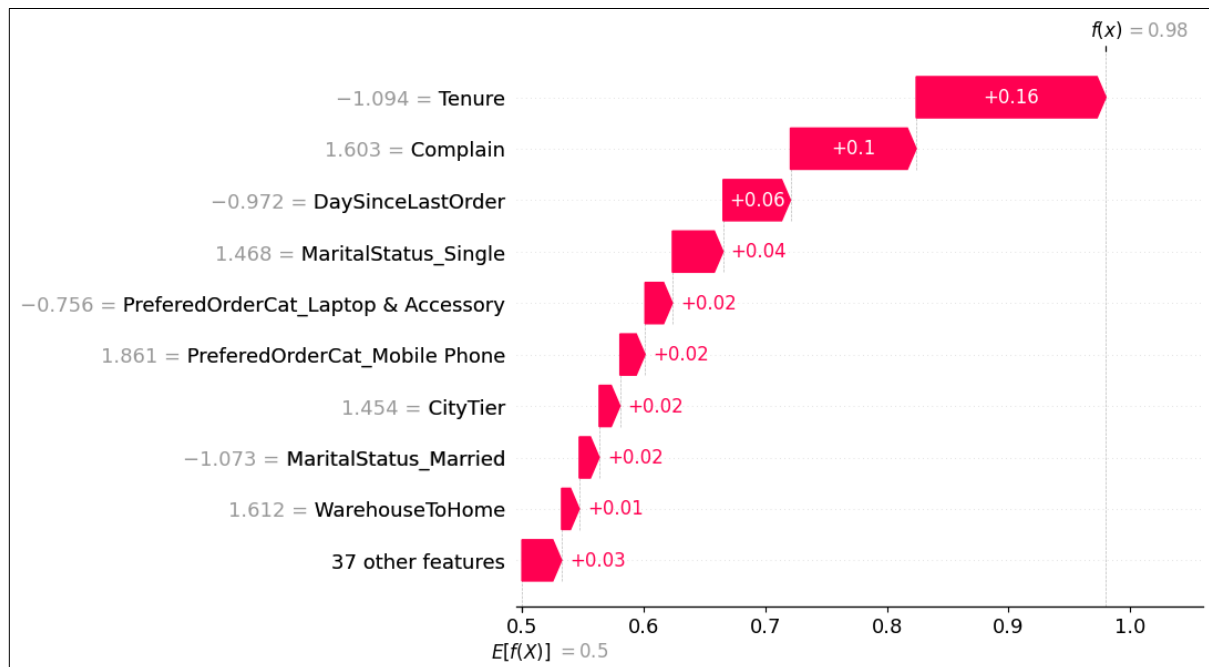


Table 5.3

Example of a Business-Readable SHAP Explanation

Customer_Index	Local_SHAP_Reason_1	Local_SHAP_Reason_2	Local_SHAP_Reason_3
0	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Customer satisfaction reduced this customer's predicted churn risk.
1	Customer tenure reduced this customer's predicted churn risk.	Customer satisfaction reduced this customer's predicted churn risk.	Complaint history increased this customer's predicted churn risk.
2	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Customer satisfaction reduced this customer's predicted churn risk.
3	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Time since last order reduced this customer's predicted churn risk.
4	Complaint history increased this customer's predicted churn risk.	Customer tenure reduced this customer's predicted churn risk.	Delivery distance reduced this customer's predicted churn risk.

The artefact successfully translated technical SHAP outputs into concise business-readable explanations describing the key behavioural factors influencing each customer's predicted churn risk. Rather than requiring managers to interpret feature contribution values, the system presented natural-language explanations that justified predictions using operational business terminology.

These explanations support interpretation of individual predictions but should not be interpreted as evidence of causal relationships.

5.3.3 Evaluation of Business-Readable Explanations

A distinguishing feature of the developed artefact is the automatic conversion of technical explainability outputs into business-oriented language suitable for non-technical users. Integrating these explanations within the Customer Profile and Recommendation Engine reduced the need for users to interpret statistical visualisations while maintaining consistency with the underlying predictive model.

Overall, the Explainability Engine successfully translated technical SHAP outputs into business-readable explanations suitable for non-technical users, providing the interpretive foundation for the Recommendation Engine.

5.4 Evaluation of Actionable Decision Support

While the Explainability Engine improves understanding of customer churn predictions, effective customer retention also requires guidance on appropriate managerial actions. This section evaluates whether the Recommendation Engine and Counterfactual Recommendation Module successfully transformed predictive outputs into practical decision support for retail SMEs.

5.4.1 Evaluation of the Recommendation Engine

The Recommendation Engine was evaluated according to its ability to integrate predictive analytics, customer behaviour segmentation, explainability, and business rules into coherent customer retention recommendations.

Table 5.4

Example output generated by the Recommendation Engine.

Customer_Index	Churn_Probability	Churn_Risk	Customer_Behavior_Segment	Priority_Score	Priority_Label	Final_Management_Action	Local_SHAP_Reason_1	Local_SHAP_Reason_2	Local_SHAP_Reason_3
0	0.005	Low Churn Risk	Inactive Customers	10.5	Monitor	No immediate costly action; monitor or include in low-cost automated campaign.	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Customer satisfaction reduced this customer's predicted churn risk.
1	0.055	Low Churn Risk	New and Developing Customers	5.5	Monitor	Continue onboarding and monitor engagement.	Customer tenure reduced this customer's predicted churn risk.	Customer satisfaction reduced this customer's predicted churn risk.	Complaint history increased this customer's predicted churn risk.
2	0.045	Low Churn Risk	New and Developing Customers	4.5	Monitor	Continue onboarding and monitor engagement.	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Customer satisfaction reduced this customer's predicted churn risk.

3	0.05	Low Churn Risk	New and Developing Customers	5.0	Monitor	Continue onboarding and monitor engagement.	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Time since last order reduced this customer's predicted churn risk.
4	0.155	Low Churn Risk	New and Developing Customers	15.5	Monitor	Continue onboarding and monitor engagement.	Complaint history increased this customer's predicted churn risk.	Customer tenure reduced this customer's predicted churn risk.	Delivery distance reduced this customer's predicted churn risk.
5	0.035	Low Churn Risk	Frequent Customers Losing Interest	18.5	Monitor	Monitor engagement trend before offering discount.	Customer tenure reduced this customer's predicted churn risk.	Cashback behaviour reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.
6	0.145	Low Churn Risk	New and Developing Customers	14.5	Monitor	Continue onboarding and monitor engagement.	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Customer satisfaction reduced this customer's predicted churn risk.
7	0.04	Low Churn Risk	New and Developing Customers	4.0	Monitor	Continue onboarding and monitor engagement.	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Time since last order reduced this customer's predicted churn risk.
8	0.13	Low Churn Risk	New and Developing Customers	13.0	Monitor	Continue onboarding and monitor engagement.	Customer tenure reduced this customer's predicted churn risk.	Complaint history increased this customer's predicted churn risk.	Cashback behaviour reduced this customer's predicted churn risk.
9	0.035	Low Churn Risk	New and Developing Customers	3.5	Monitor	Continue onboarding and monitor engagement.	Customer tenure reduced this customer's predicted churn risk.	Cashback behaviour reduced this customer's predicted churn risk.	Complaint history increased this customer's predicted churn risk.
10	0.045	Low Churn Risk	Previously Valuable Customers	24.5	Monitor	Monitor behaviour and use light-touch reactivation message.	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Time since last order reduced this customer's predicted churn risk.
11	0.785	High Churn Risk	New and Developing Customers	78.5	Immediate Action	Provide onboarding support and first-purchase retention incentive.	Customer tenure increased this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Customer satisfaction increased this customer's predicted churn risk.
12	0.04	Low Churn Risk	New and Developing Customers	4.0	Monitor	Continue onboarding and monitor engagement.	Customer tenure reduced this customer's predicted churn risk.	Complaint history reduced this customer's predicted churn risk.	Time since last order reduced this customer's predicted churn risk.

The generated recommendations combined churn probability, RFM customer segmentation, retention priority, and business-readable SHAP explanations to provide managers with customer-specific retention guidance. Rather than presenting churn predictions in isolation, the system contextualised each prediction using customer value and behavioural characteristics, enabling more informed prioritisation of retention activities.

An important outcome of this evaluation was that customers with similar churn probabilities did not necessarily receive identical recommendations. By incorporating customer value and behavioural segmentation, the Recommendation Engine supported more strategic allocation of limited retention resources than approaches based solely on predictive probability. The inclusion of business-readable explanations further strengthened transparency by allowing managers to understand the analytical reasoning supporting each recommendation.

Although the current implementation relies on predefined business rules, the evaluation demonstrates that this approach provides consistent and interpretable recommendations suitable for prototype-level decision support. Future implementations could extend this capability by incorporating adaptive recommendation techniques that learn from historical intervention outcomes.

5.4.2 Evaluation of Counterfactual Recommendations

The Counterfactual Recommendation Module was evaluated to determine whether it generated feasible behavioural changes capable of supporting customer intervention planning.

Figure 5.6

Example counterfactual explanation generated using the DiCE framework.

Customer_Index	Original_Probability	Target_Probability	Feature_To_Change	Current_Value_Scaled	Suggested_Value_Scaled	Status
0	11	0.785	0.185	Tenure	-1.213	1.244 Success
1	41	0.735	0.155	Tenure	-1.094	5.683 Success
2	53	0.800	0.410	Tenure	-1.213	-0.408 Success
3	53	0.800	0.410	CashbackAmount	-0.402	-1.293 Success
4	59	0.850	0.345	Tenure	-1.213	5.615 Success

Table 5.5

Example of a business-readable counterfactual recommendation

Customer_Index	Original_Probability	Target_Probability	Feature_To_Change	Current_Value_Scaled	Suggested_Value_Scaled	Status	Counterfactual_Manager_Action
11	0.785	0.185	Tenure	-1.213	1.244	Success	Review customer behaviour and apply suitable retention action.
41	0.735	0.155	Tenure	-1.094	5.683	Success	Review customer behaviour and apply suitable retention action.
53	0.8	0.41	Tenure	-1.213	-0.408	Success	Review customer behaviour and apply suitable retention action.
53	0.8	0.41	CashbackAmount	-0.402	-1.293	Success	Offer cashback or loyalty reward to encourage retention.

59	0.85	0.345	Tenure	-1.213	5.615	Succe ss	Review customer behaviour and apply suitable retention action.
59	0.85	0.345	DaySinceLastO rder	0.997	3.98	Succe ss	Encourage the customer to place a new order sooner through a re- engagement campaign.
79	0.76	0.19	Tenure	-1.213	3.741	Succe ss	Review customer behaviour and apply suitable retention action.
81	0.745	0.33	Tenure	-1.213	0.523	Succe ss	Review customer behaviour and apply suitable retention action.

Unlike SHAP explanations, which explain why a prediction was generated, counterfactual recommendations identify behavioural changes associated with a lower predicted churn risk. Within the developed artefact, these recommendations focused on modifiable customer characteristics such as purchasing behaviour, engagement, and transaction recency, thereby supporting practical retention planning.

5.4.3 Overall Evaluation of Actionable Decision Support

The combined evaluation demonstrates that the developed artefact extends conventional customer churn prediction by integrating prediction, explanation, prioritisation, and intervention guidance within a single decision support environment. Rather than requiring managers to interpret multiple analytical outputs independently, the system presents integrated recommendations that support transparent and evidence-based customer retention planning.

From a Business Informatics perspective, this integration represents a key contribution of the research. User evaluation indicated that participants found the combined presentation of predictions, explanations, and recommendations easier to understand than predictive outputs alone, suggesting that explainability provides greatest value when embedded within a broader decision support workflow.

5.5 Evaluation of the Integrated Decision Support Interface

5.5.1 Functional Evaluation of the Decision Support Interface

The Integrated Decision Support Interface was evaluated to determine whether it effectively presented analytical outputs generated throughout the implementation in a manner that supports managerial decision-making. The interface supports organisation-level monitoring and customer-level decision-making by integrating analytical outputs within a unified workflow.

A key strength of the interface is the integration of multiple analytical outputs into a unified workflow. Rather than requiring managers to interpret prediction results, explainability outputs, and recommendations separately, the interface presents these components within a single contextual view. This integrated presentation may reduce cognitive effort, improve interpretability, and support evidence-based decision-making by enabling managers to understand which customers require intervention, why they have been prioritised, and what actions may reduce churn risk.

From a Business Intelligence perspective, the interface successfully bridges the gap between advanced machine learning techniques and practical organisational decision-making. By translating complex analytical outputs into business-oriented information, the developed system improves accessibility for non-technical stakeholders while maintaining transparency regarding the reasoning underlying predictive recommendations. Consequently, the interface functions as the operational component of the developed decision support system rather than simply acting as a visualisation layer.

5.5.2 User Evaluation

A user evaluation was conducted to assess the usability, interpretability, and practical usefulness of the developed decision support prototype. The evaluation involved eleven participants, including retail managers, business students, a small business owner, and a data engineer. This participant mix was appropriate for a formative Design Science Research evaluation because the objective was to assess the prototype's usability, interpretability, and practical value rather than to produce statistically generalisable findings.

Participants were first introduced to the purpose of the prototype and then asked to review the dashboard functions, including customer churn prediction, customer segmentation, retention priority, SHAP explanations, and counterfactual recommendations. After interacting with the prototype, participants completed a Microsoft Forms questionnaire containing Likert-scale questions and open-ended feedback items. The complete evaluation questionnaire administered through Microsoft Forms is included in Appendix E.

Table 5.6
Participant Profile

Participant Type	Frequency
Retail Manager	5
Business Student	4
Small Business Owner	1
Data Engineer	1
Total	11

The participant group included business-oriented and technically informed users with varying levels of dashboard experience, providing diverse perspectives during the formative evaluation. A detailed summary of participant roles and prior familiarity with business analytics dashboards is presented in Appendix F.

Table 5.7

Summary of User Evaluation Results (n = 11)

Evaluation Statement	Mean Score / 5
The dashboard was easy to navigate	4.27
The dashboard helped identify customers requiring retention action	4.18
SHAP explanations improved understanding of churn predictions	4.27
Counterfactual recommendations were useful	4.0
I would use this type of dashboard for customer retention decision-making	4.36

The quantitative findings indicate a positive overall response to the developed prototype. The highest mean scores were recorded for ease of navigation and willingness to use the dashboard for customer retention decision-making, both achieving a mean score of 4.27 and 4.36 respectively out of 5. This suggests that participants generally perceived the interface as accessible and practically relevant. The SHAP explanation score of 4.27 indicates that participants found the explainability component useful for understanding churn predictions, while the counterfactual recommendation score of 4.0 suggests that participants considered the actionable recommendations valuable, although slightly less strongly than the navigation and prediction features.

The qualitative responses were analysed thematically to identify recurring patterns in participant feedback. Four main themes emerged: integrated customer information, explainability and understanding, ease of use, and suggested future improvements.

Table 5.8
Qualitative Themes from User Evaluation

Theme	Summary of Feedback	Example Participant Comment
Integrated customer information	Participants valued the Customer Profile and Retention Priority Centre because they combined customer details, churn risk, and recommended actions.	“Customer Profile gives a clear idea about every customer.”
Explainability and understanding	Participants indicated that explanations and churn drivers helped them understand why customers were predicted to churn.	“The dashboard provides a clear representation... there is little scope for confusion.”
Ease of use	Participants generally found the dashboard straightforward and easy to navigate.	“It looks quite straightforward.”
Future improvements	Suggestions focused on layout refinement, real-time data, improved filtering, and more personalised recommendations.	“Make it more user friendly and add filters.”

The qualitative feedback supports the quantitative findings by showing that participants valued the integrated nature of the dashboard. In particular, the Customer Profile and Retention Priority Centre were frequently identified as useful because they brought together customer risk, priority, explanation, and recommended actions in one location. This finding supports the design decision to present predictive analytics, explainability, and recommendations within a single decision support interface rather than as separate outputs. The complete anonymous participant responses that informed the thematic analysis are reproduced in Appendix G.

The feedback also suggests that explainability improved participant understanding of the model outputs. Participants recognised the value of seeing churn drivers and recommendations alongside prediction results, indicating that the SHAP-based explanations and business-readable outputs supported transparency.

Participants also identified opportunities for future enhancement. Suggested improvements included simplifying the layout further, improving filtering options, incorporating real-time data, and expanding the personalisation of recommendations. These suggestions are consistent with the prototype nature of the system and provide useful direction for future development.

Overall, the user evaluation provides evidence that the developed prototype is understandable, useful, and relevant for customer retention decision-making. Although the evaluation involved a relatively small sample of eleven participants, it provided valuable formative feedback from both business-oriented and technically informed users. The findings therefore offer evidence of the prototype's perceived usability and practical usefulness but should not be interpreted as statistically generalisable across the retail SME sector.

Critical limitation of the evaluation

The evaluation was formative and involved a purposive sample of eleven participants rather than a statistically representative population. The findings therefore provide evidence of perceived usability, interpretability, and practical usefulness but cannot establish long-term organisational effectiveness or generalisability across the retail SME sector. Nevertheless, the sample included both business and technically informed participants, making the feedback appropriate for evaluating the prototype during the Design Science Research process.

5.6 Overall Artefact Evaluation

The Design Science Research methodology evaluates an artefact according to its ability to address the identified organisational problem. Accordingly, the developed artefact was evaluated using evidence from predictive performance, explainability, actionable recommendations, decision support functionality, and user evaluation. Collectively, these findings indicate that the artefact successfully achieved the research objectives established in Chapter 1 and demonstrates the feasibility of integrating Explainable Artificial Intelligence within a Business Intelligence decision support environment for customer retention.

From a Business Informatics perspective, the artefact contributes by combining customer churn prediction, behavioural segmentation, SHAP explainability, counterfactual recommendations, and an interactive dashboard within a unified workflow. Rather than providing churn predictions alone, the system supports customer prioritisation, interpretation of model outputs, and planning of retention actions through an integrated decision support process. The technical evaluation demonstrated reliable predictive performance, while the user evaluation indicated positive perceptions of usability, interpretability, and practical usefulness within the scope of the prototype.

Table 5.9
Research Objective Achievement Matrix

Research Objective	Evidence Presented	Chapter/Section	Achievement
Objective 1: Investigate the application of Business Intelligence, machine learning, and Explainable Artificial Intelligence (XAI) techniques for customer retention in retail SMEs.	Comprehensive literature review established the theoretical foundation for Business Intelligence, customer churn prediction, Explainable AI, RFM segmentation, and decision support systems.	Chapter 2	Achieved
Objective 2: Design and develop an integrated decision support system combining customer churn prediction,	A functional Streamlit-based decision support system integrating predictive analytics, behavioural segmentation,	Chapters 3–4	Achieved

RFM segmentation, SHAP explainability, and counterfactual recommendations.	explainability, and recommendation generation was successfully implemented.		
Objective 3: Evaluate the predictive performance, explainability, and practical usefulness of the developed prototype using technical performance measures and user-centred evaluation.	The Random Forest model, SHAP explainability, counterfactual recommendations, dashboard functionality, and user evaluation demonstrated the technical and practical effectiveness of the developed artefact.	Chapter 5	Achieved
Objective 4: Examine the ethical, cultural, and governance considerations associated with the responsible adoption of AI-enabled decision support systems.	Ethical, cultural, governance, and human-centred AI considerations were critically analysed, including transparency, privacy, fairness, Māori Data Sovereignty, and responsible AI principles.	Chapter 6	Achieved

Overall, the evaluation demonstrates that the developed artefact satisfies the defined research objectives and provides a technically sound and practically useful prototype for Explainable AI-enabled customer retention decision support. Although further validation within operational retail environments is required, the findings indicate that integrating predictive analytics with explainability and decision support can enhance the practical application of customer churn analysis in retail SMEs.

5.7 Chapter Summary

This chapter evaluated the developed artefact using predictive performance, explainability, actionable recommendations, dashboard functionality, and user evaluation. The findings demonstrate that the prototype achieved the research objectives and provides a technically sound and practically useful foundation for Explainable AI-enabled customer retention decision support. Chapter 6 critically reflects on the ethical considerations, limitations, and future research opportunities.

Chapter 6: Critical Reflection

6.1 Introduction

This chapter critically reflects on the developed Explainable Artificial Intelligence-enabled Business Intelligence decision support system. It first examines the ethical, cultural, and governance responsibilities associated with responsible organisational use of the artefact. It then evaluates the technical and practical limitations identified through implementation and evaluation and outlines corresponding opportunities for future research and operational development.

6.2 Ethical and Cultural Considerations

The implementation of Artificial Intelligence-enabled decision support systems requires consideration of ethical, governance, and cultural responsibilities alongside predictive performance. Although the developed artefact demonstrates the practical value of integrating Business Intelligence, Explainable Artificial Intelligence (XAI), and customer retention analytics, responsible organisational deployment depends on transparency, fairness, privacy, accountability, and meaningful human oversight rather than predictive accuracy alone (National Institute of Standards and Technology, 2023; Organisation for Economic Co-operation and Development, 2019).

A key ethical strength of the developed artefact is the integration of SHAP explanations and counterfactual recommendations, which improve the transparency of customer churn predictions by providing managers with interpretable evidence to support decision-making. Consistent with the objectives of Explainable Artificial Intelligence, these techniques enable business users to understand both the factors contributing to customer churn predictions and the potential actions that may improve retention outcomes (Arrieta et al., 2019; Lundberg & Lee, 2017). However, explainability alone does not eliminate ethical risks. Predictive outputs remain dependent on the quality and representativeness of historical customer data, meaning that incomplete or biased datasets may influence future predictions and recommendations. Organisations should therefore implement appropriate data governance practices, regularly monitor predictive outcomes, and evaluate model performance across diverse customer groups to minimise unintended bias (National Institute of Standards and Technology, 2023).

The developed prototype was intentionally designed to support human-centred decision-making rather than automated customer management. Predictions, explanations, and recommendations provide decision-support information, while final customer retention decisions remain the responsibility of managers, who must also consider organisational objectives, customer relationships, and contextual factors beyond those represented in the predictive model. This approach aligns with responsible AI principles that emphasise human oversight, accountability, and organisational responsibility when deploying AI-assisted decision support systems (Organisation for Economic Co-operation and Development, 2019).

The artefact also incorporates a design decision intended to improve both fairness and practical usability. Although the underlying Random Forest model utilises all selected predictive features during classification, the user-facing SHAP explanations and counterfactual recommendations intentionally exclude immutable demographic attributes, such as gender and marital status. Instead, the dashboard presents business-actionable behavioural factors that

managers can realistically influence through retention interventions. This design encourages intervention-oriented decision-making while reducing the likelihood that recommendations are interpreted on the basis of immutable personal characteristics (Arrieta et al., 2019).

From a privacy perspective, the prototype was developed using an anonymised customer dataset and focused exclusively on analytical variables required for churn prediction and customer segmentation. Nevertheless, operational deployment would require compliance with applicable privacy legislation, secure data management practices, and appropriate governance over the collection, storage, processing, and retention of customer information. Regular auditing of data quality, model performance, and access controls would also be necessary to ensure the continued responsible use of customer data (National Institute of Standards and Technology, 2023).

Within the New Zealand context, future implementation should also recognise broader cultural responsibilities relating to Māori Data Sovereignty. Although Māori data were not specifically represented in the dataset used for this research, future deployment should consider the CARE Principles for Indigenous Data Governance and the guidance provided by Te Mana Rauunga to promote collective benefit, authority to control, responsibility, and ethical stewardship of data. Incorporating these principles would strengthen culturally responsive governance and support greater organisational trust in AI-enabled decision support systems (Global Indigenous Data Alliance, 2019; Te Mana Rauunga, n.d.).

Overall, the artefact demonstrates that responsible AI-enabled decision support requires more than predictive performance. Integrating explainability with human oversight, privacy, governance, and culturally informed data practices provides a stronger foundation for responsible organisational adoption.

6.3 Limitations and Future Research Directions

This section critically examines the limitations of the developed artefact and identifies corresponding opportunities for further research. The limitations are considered from two complementary perspectives: technical constraints affecting the analytical and computational components of the system, and practical constraints affecting organisational implementation, adoption, and evaluation.

6.3.1 Technical Limitations

Although the developed decision support system demonstrated promising analytical capabilities, several technical limitations should be acknowledged when interpreting the findings. The first limitation relates to the dataset used throughout the research. The predictive model was developed and evaluated using a single historical customer dataset that, while appropriate for demonstrating the proposed approach, may not fully represent the diversity of customer behaviours observed across different retail organisations. Variations in customer demographics, purchasing behaviour, market conditions, and organisational practices may influence model performance when applied to alternative business environments.

The predictive model was trained using historical customer data and therefore reflects behavioural patterns present during model development. The technical evaluation demonstrated strong predictive performance using the available historical dataset; however,

maintaining comparable performance in operational environments would require periodic retraining using updated customer data to address model drift.

The comparative evaluation used a common preprocessing pipeline to ensure consistency; however, this may not represent the optimal algorithm-specific configuration, particularly for models such as CatBoost that can natively process categorical variables.

Another limitation relates to the implementation of the RFM segmentation. Because the selected dataset did not include a direct measure of customer monetary expenditure, CashbackAmount was used as a proxy for the Monetary component. Although this enabled consistent behavioural segmentation, cashback may not fully represent actual customer value. Future research should validate the proposed framework using datasets containing complete transaction value information to strengthen the accuracy of behavioural segmentation. The operationalisation of the adapted RFM framework and the use of CashbackAmount as a proxy for the Monetary dimension are documented in Appendix C.

SHAP explanations improve transparency by explaining model behaviour but should not be interpreted as evidence of causal relationships. Likewise, counterfactual recommendations identify feasible behavioural changes associated with alternative prediction outcomes rather than guaranteeing organisational success. These outputs should therefore complement, rather than replace, managerial judgement.

The technical evaluation also highlighted several prototype-level limitations. Although the Recommendation Engine successfully generated consistent and interpretable customer retention guidance, it relied on predefined business rules rather than adaptive learning from historical intervention outcomes. Furthermore, the user evaluation identified opportunities to improve filtering functionality, recommendation personalisation, and integration with real-time organisational data. These findings indicate that the artefact demonstrates prototype-level technical feasibility but requires further refinement and validation before operational deployment.

6.3.2 Practical Limitations and Future Research Directions

Although the artefact demonstrates the feasibility of integrating Business Intelligence, Explainable Artificial Intelligence, and decision support within a unified customer retention system, several practical considerations should be addressed before deployment in operational retail environments.

First, the prototype was developed using a static historical dataset within a Streamlit environment. Operational implementation would require integration with enterprise information systems, including Customer Relationship Management (CRM) platforms, transactional databases, and automated data pipelines to ensure that customer information remains current and predictions reflect evolving customer behaviour.

Second, successful organisational adoption depends on more than predictive performance. Effective deployment requires user training, organisational trust, governance procedures, and clearly defined decision-making responsibilities to ensure that managers can interpret predictive outputs and apply recommendations appropriately. Furthermore, organisational adoption depends on integrating analytical outputs into existing decision-making processes rather than treating AI recommendations as standalone decisions. This reinforces the role of the artefact as a decision-support tool rather than an autonomous decision-making system.

Third, maintaining long-term performance would require automated retraining, calibration assessment, and continuous monitoring of predictive accuracy and fairness. This would improve the long-term reliability of the decision support system.

Future research could also extend the Recommendation Engine by incorporating adaptive recommendation techniques that learn from historical intervention outcomes rather than relying solely on predefined business rules. Integrating additional customer information, such as customer satisfaction, marketing engagement, or digital interaction data, may further improve prediction quality and recommendation relevance.

Finally, longitudinal evaluation within operational retail environments would provide stronger evidence of organisational impact by assessing customer retention outcomes, user adoption, marketing effectiveness, and business value over extended periods. Such studies would provide a more comprehensive assessment of the practical benefits of Explainable Artificial Intelligence-enabled Business Intelligence decision support systems.

Chapter 7: Discussion

7.1 Introduction

This chapter interprets the findings of the research in relation to the objectives established in Chapter 1 and the literature reviewed in Chapter 2. Rather than presenting additional results, it critically discusses the significance of the developed artefact for Business Intelligence, Explainable Artificial Intelligence (XAI), and customer retention decision support within retail SMEs. The chapter synthesises the evaluation findings, identifies the research contributions, and considers the practical implications of the proposed approach from a Business Informatics perspective before concluding the study.

7.2 Discussion of Findings

The findings indicate that the organisational value of Artificial Intelligence for customer retention extends beyond predictive performance alone. Although the developed Random Forest model demonstrated strong predictive capability, the evaluation suggests that prediction accuracy by itself is insufficient to support effective managerial decision-making. Consistent with the literature reviewed in Chapter 2, business users require analytical outputs that are understandable, interpretable, and capable of informing appropriate retention actions rather than simply identifying customers at risk of churn. The research therefore supports the view that the practical value of AI-enabled Business Intelligence depends not only on model performance but also on its ability to support human decision-making through transparent and actionable insights.

A key finding of this research is that explainability becomes more valuable when embedded within a broader decision support process rather than being implemented as an isolated analytical capability. While previous studies have demonstrated the technical effectiveness of SHAP explanations or counterfactual reasoning independently, the developed artefact illustrates how these techniques can complement predictive analytics by providing both an explanation of why a customer has been classified as high risk and guidance on potential intervention strategies. The evaluation therefore suggests that explainability contributes most effectively when integrated into an operational workflow that supports customer prioritisation, interpretation, and action planning.

The findings also reinforce the evolving role of Business Intelligence within organisational decision-making. Traditional Business Intelligence systems primarily support descriptive and diagnostic analysis by reporting historical business performance. In contrast, the developed artefact demonstrates how Business Intelligence can be extended through predictive analytics, explainability, and recommendation generation to support prospective decision-making. Rather than replacing managerial judgement, the system provides evidence that assists managers in identifying at-risk customers, interpreting prediction outcomes, and evaluating potential retention strategies. This progression aligns with contemporary Business Informatics research, which increasingly positions Business Intelligence as an integrated decision support capability rather than solely a reporting function.

The user evaluation further supports the practical relevance of this integrated approach. Participants generally reported that combining customer profiles, churn predictions, business-readable explanations, and retention recommendations within a single interface improved their

understanding of analytical outputs and made the system easier to use. Although the evaluation was formative and involved a relatively small purposive sample, these findings suggest that integrating explainability within the decision support workflow can improve the perceived usability and interpretability of AI-enabled Business Intelligence systems. This observation is consistent with previous research emphasising that organisational adoption of AI depends not only on predictive capability but also on user trust, transparency, and ease of interpretation.

Taken together, the findings suggest that the principal contribution of the research lies not in the individual analytical techniques employed, but in demonstrating how predictive analytics, behavioural segmentation, explainability, and decision support can be integrated into a coherent Business Intelligence workflow. The developed artefact therefore extends existing customer churn research by illustrating that the practical value of Explainable Artificial Intelligence is realised most effectively when technical explanations are translated into decision-support information that managers can readily understand and apply within organisational contexts.

7.3 Research Contributions

This research contributes to the fields of Business Informatics and Explainable Artificial Intelligence through three complementary contributions encompassing knowledge, artefact development, and practical application.

First, the research contributes to knowledge by demonstrating that the practical value of Explainable Artificial Intelligence is enhanced when explainability is embedded within an integrated Business Intelligence decision support workflow rather than applied as an isolated analytical technique. While previous customer churn studies have primarily focused on improving predictive performance or evaluating individual explainability methods, the findings suggest that combining prediction, behavioural segmentation, explanation, and recommendation within a unified environment provides greater support for managerial interpretation and decision-making. This extends existing research by emphasising the importance of operational integration alongside predictive capability.

Second, the research contributes through the Design Science Research artefact developed during the study. The implemented prototype integrates customer churn prediction, RFM customer segmentation, SHAP explainability, counterfactual recommendations, and an interactive Streamlit dashboard within a single decision support environment. The artefact demonstrates the technical feasibility of combining these complementary analytical capabilities while providing a reusable implementation framework that may inform future Explainable AI-enabled Business Intelligence applications.

Third, the research provides a practical contribution for retail SMEs by illustrating how advanced analytical techniques can be translated into business-oriented decision support. The incorporation of business-readable explanations, customer prioritisation, and actionable recommendations reduces the technical complexity associated with machine learning models and supports their accessibility for non-technical decision-makers. Although further validation within operational environments is required, the findings indicate that the proposed approach has the potential to improve the practical usability of AI-enabled customer retention systems.

7.4 Business Implications

The findings of this research have several practical implications for retail SMEs seeking to strengthen customer retention through data-driven decision-making. Many SMEs continue to rely on managerial experience, intuition, or descriptive reports when identifying customers at risk of churn. The developed artefact demonstrates how AI-enabled Business Intelligence can complement, rather than replace, managerial judgement by providing predictive insights that are supported by interpretable explanations and actionable recommendations. This reinforces the role of Business Intelligence as a decision support capability that enhances human decision-making instead of automating it.

A key implication concerns the prioritisation of limited organisational resources. Customer retention initiatives often involve financial and operational constraints that require managers to determine which customers should receive targeted interventions. By combining churn probability with RFM-based customer value and retention priority, the artefact provides a structured approach for identifying customers who are both strategically valuable and at greater risk of churn. This enables retention efforts to be prioritised more systematically and supports more effective allocation of marketing resources.

The findings also highlight that successful organisational adoption of Artificial Intelligence depends on more than predictive performance. The evaluation suggests that business users place value on analytical outputs that they can understand, interpret, and relate to existing decision-making processes. Presenting SHAP explanations and counterfactual recommendations in business-readable language reduces the technical complexity of machine learning outputs and supports communication between technical specialists and business stakeholders. Consequently, explainability should be considered an important design consideration for AI-enabled decision support systems rather than an optional technical enhancement.

From a broader Business Informatics perspective, the research illustrates how Business Intelligence systems can evolve from supporting descriptive reporting towards enabling predictive and prescriptive decision support. Integrating predictive analytics with explainability and recommendation generation enables organisations to move beyond understanding historical customer behaviour and towards evaluating potential future interventions. This demonstrates how AI technologies can be incorporated into Business Intelligence environments to improve organisational responsiveness while maintaining human oversight of business decisions.

Although the framework was developed and evaluated within the context of retail SMEs, its modular design suggests that similar principles could be adapted to other customer-centric industries, such as banking, telecommunications, or subscription-based services, where customer retention is a strategic priority. However, because this research evaluated the artefact only within the retail domain, the applicability of the framework to other sectors remains a direction for future investigation rather than a confirmed outcome.

Overall, the findings indicate that the organisational value of AI-enabled Business Intelligence is realised not simply through accurate prediction, but through the delivery of decision-support information that managers can confidently interpret and apply. Combining predictive capability with explainability and actionable recommendations provides a practical foundation for more transparent, evidence-informed, and user-centred customer retention decision-making.

7.5 Chapter Summary

This chapter interpreted the research findings in relation to the study objectives and the existing literature on Business Intelligence, Explainable Artificial Intelligence, and customer churn prediction. The discussion demonstrated that the primary contribution of the research lies in integrating predictive analytics, explainability, and decision support within a unified Business Intelligence environment that supports transparent and actionable customer retention decisions. The chapter also outlined the research contributions and highlighted the practical implications of the proposed approach for retail SMEs. The final chapter concludes the dissertation by revisiting the research objectives, summarising the overall findings, acknowledging the study limitations, and identifying directions for future research.

Chapter 8: Conclusion

This research aimed to design, develop, and evaluate an Explainable Artificial Intelligence (XAI)-enabled Business Intelligence decision support system to support customer retention decision-making in retail SMEs. Guided by the Design Science Research methodology, the study produced and evaluated a functional artefact that integrates customer churn prediction, RFM customer segmentation, SHAP explainability, counterfactual recommendations, and an interactive Streamlit dashboard within a unified Business Intelligence decision support environment.

The evaluation findings demonstrate that the research aim and objectives were successfully achieved. The developed artefact provided reliable customer churn prediction while supporting managerial interpretation through business-readable SHAP explanations and actionable counterfactual recommendations. By integrating predictive analytics with customer segmentation and interactive visualisation, the system enabled users to identify customers at risk of churn, understand the key factors influencing individual predictions, and evaluate appropriate retention strategies within a single decision support workflow. Collectively, these findings indicate that the organisational value of Explainable Artificial Intelligence extends beyond predictive performance by improving the transparency, interpretability, and practical usefulness of Business Intelligence for customer retention decision-making.

Accordingly, the evaluation confirms that the study achieved its objectives of developing a customer churn prediction model, incorporating explainability through SHAP and counterfactual reasoning, integrating these capabilities within a Business Intelligence decision support artefact, and evaluating its technical and practical effectiveness.

The evaluation findings also provide clear answers to the three research questions established in Chapter 1. First, the findings demonstrate that SHAP-based explainability improved the interpretability of customer churn predictions by translating complex machine learning outputs into business-readable explanations, thereby supporting managerial understanding of the factors influencing individual predictions. Second, the findings indicate that counterfactual explanations extended predictive analytics by identifying feasible behavioural changes that informed actionable customer retention recommendations, enabling managers to move from prediction towards intervention planning. Third, the evaluation demonstrates that integrating predictive analytics, customer segmentation, explainability, and recommendation generation within a unified Business Intelligence decision support environment provides more effective support for customer retention decision-making than predictive analytics alone. Collectively, these findings confirm that the research questions were successfully addressed through the design, implementation, and evaluation of the developed artefact.

Beyond addressing the research questions, the study contributes to the growing body of Business Informatics research by demonstrating that the practical value of Artificial Intelligence lies not only in accurate prediction but also in its integration within organisational decision support processes. The research illustrates how predictive analytics, explainability, and business-oriented recommendations can operate as complementary capabilities within a unified decision support environment that supports managerial judgement rather than prediction alone. In doing so, the study addresses a recognised gap in the customer churn literature, where research has frequently prioritised predictive performance over interpretability and practical business application.

Although the artefact was developed and evaluated using a publicly available retail dataset and a limited user evaluation, the findings provide a foundation for future research involving enterprise deployment, real-time data integration, automated model monitoring and retraining, and longitudinal evaluation within operational retail environments. Such research would provide stronger evidence regarding organisational adoption, long-term business impact, and the sustained effectiveness of Explainable Artificial Intelligence-enabled Business Intelligence systems under changing business conditions.

Overall, this research demonstrates that effective customer retention requires more than accurate prediction. The findings indicate that integrating predictive analytics with explainability, customer behaviour analysis, and actionable decision support enables Artificial Intelligence to become a transparent, interpretable, and trustworthy managerial capability rather than a standalone predictive technology. By combining these capabilities within a unified Business Intelligence decision support environment, the developed artefact contributes both a practical solution for retail SMEs and empirical evidence supporting the role of Explainable Artificial Intelligence in improving organisational decision-making. The research therefore demonstrates how Explainable Artificial Intelligence can move beyond enhancing machine learning transparency to become an integral component of human-centred Business Intelligence for customer retention.

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Appendices

The following appendices provide supplementary material supporting the design, implementation, and evaluation of the developed Explainable AI-enabled Business Intelligence decision support system. They are intended to complement, rather than replace, the evidence and discussion presented in the main body of the dissertation.

Appendix A: Technical Overview of the Developed Research Artefact

This appendix provides a technical overview of the developed Explainable Artificial Intelligence (XAI)-enabled Business Intelligence decision support artefact. It summarises the principal technologies, analytical components, development environment, and deployment platform implemented during the research. The overview provides a concise technical reference that complements the detailed design and implementation presented in Chapters 3 and 4.

Table A.1

Technical Overview of the Developed Research Artefact

Component	Implementation
Research Methodology	Design Science Research (DSR)
Artefact Type	Explainable AI-enabled Business Intelligence Decision Support System
Application Domain	Customer Retention Decision Support for Retail SMEs
Programming Language	Python (Version 3.14.0)
Development Environment	Jupyter Notebook and Visual Studio Code
Dashboard Framework	Streamlit (Version 1.58.0)
Deployment Platform	Streamlit Community Cloud
Source Code Repository	GitHub Repository
Machine Learning Framework	Scikit-learn
Selected Predictive Model	Random Forest Classifier (selected following comparative evaluation of five supervised machine learning algorithms)
Explainability Technique	SHAP (SHapley Additive exPlanations)
Counterfactual Explanation Framework	DiCE (Diverse Counterfactual Explanations)
Customer Behaviour Analysis	Recency–Frequency–Monetary (RFM) Segmentation
Recommendation Strategy	Rule-based recommendation engine integrating churn probability, RFM customer segmentation, SHAP explanations, and counterfactual recommendations
Data Processing Libraries	Pandas, NumPy
Visualisation Libraries	Matplotlib and Plotly
Model Persistence	Joblib
Dataset	Kaggle E-Commerce Customer Churn Dataset (Verma, n.d.)
Deployment Architecture	Modular Business Intelligence Decision Support Architecture
Evaluation Approach	Technical evaluation and formative user-centred evaluation
User Evaluation Participants	11 participants (retail managers, business students, small business owner, and data engineer)

Note.The developed artefact integrates customer churn prediction, behavioural customer segmentation, Explainable Artificial Intelligence, and decision support within a unified Business Intelligence environment. The implementation follows a modular architecture that separates data preparation, predictive modelling, explainability, recommendation generation, and user interaction while enabling these components to operate together as a coherent decision support system. The prototype was developed locally using Python and deployed through Streamlit Community Cloud to support interactive demonstration and evaluation.

Table A.1 summarises the principal technical components that constitute the developed research artefact. Rather than representing independent analytical modules, these technologies were integrated to support a complete Business Intelligence decision support workflow, enabling customer churn prediction, behavioural segmentation, explainability, recommendation generation, and interactive visualisation within a single system.

Appendix B: Design Decisions and Implementation Justifications

This appendix summarises the principal methodological and technical decisions made during the development of the research artefact and provides concise justifications for each selected approach. Rather than repeating the implementation procedures described in Chapters 3 and 4, this appendix documents the rationale underpinning the key design decisions, demonstrating how each choice contributed to achieving the research objectives and supporting the development of an Explainable Artificial Intelligence (XAI)-enabled Business Intelligence decision support system. The appendix also improves the transparency and reproducibility of the research by explicitly linking implementation decisions to the corresponding evidence presented in the dissertation.

Table B.1

Design Decisions and Implementation Justifications

Decision Category	Alternatives Considered	Selected Approach	Evidence Source	Implementation Justification
Research methodology	Case Study, Action Research, CRISP-DM	Design Science Research	Chapter 3 – Research Methodology	Design Science Research was selected because the research focused on designing, developing, and evaluating a novel decision support artefact rather than testing an existing theory.
Programming language	R, Java	Python	Chapter 4 – System Development	Python provides a comprehensive ecosystem of machine learning, explainability, data analysis, and dashboard development libraries, enabling efficient implementation of the integrated artefact.
Dashboard framework	Dash, Flask, Django	Streamlit	Chapter 4 – Dashboard Development	Streamlit enabled rapid development of an interactive web application with seamless integration of machine learning models and visual analytics while minimising software engineering complexity.
Data preprocessing	No feature scaling	StandardScaler	Chapter 4 – Data Preprocessing	Feature scaling standardised numerical variables to improve consistency across the machine learning pipeline and support

				reliable counterfactual generation.
Missing value treatment	Mean imputation, record deletion	Median imputation	Chapter 4 – Data Preprocessing	Median imputation reduced the influence of outliers commonly present in customer transaction data while preserving all available observations for analysis.
Categorical variable encoding	Label Encoding	One-Hot Encoding	Chapter 4 – Data Preprocessing	One-Hot Encoding avoided introducing artificial ordinal relationships between categorical variables and provided an appropriate representation for the modelling workflow.
Class imbalance handling	Class weighting, undersampling	SMOTE	Chapter 4 – Data Preprocessing	SMOTE was applied only to the training dataset to improve minority class representation while maintaining an unbiased evaluation using the original test data.
Predictive model	Logistic Regression, XGBoost, CatBoost, Soft Voting Ensemble	Random Forest	Chapter 5 – Model Evaluation	Random Forest was selected following comparative evaluation because it demonstrated strong predictive performance while supporting efficient TreeSHAP explainability and robust performance on structured customer data.
Explainability technique	LIME, Permutation Importance	SHAP (TreeExplainer)	Chapter 4 – Explainable AI Implementation	SHAP provides theoretically grounded feature attribution at both the global and individual prediction levels, improving transparency and enabling business users to interpret customer churn predictions.
Counterfactual explanation	Feature importance only	DiCE	Chapter 4 – Counterfactual Recommendation Module	DiCE generates actionable counterfactual recommendations by identifying minimal changes to customer characteristics that may

				reduce predicted churn risk, supporting intervention planning rather than explanation alone.
Customer segmentation	K-Means clustering, demographic segmentation	RFM Segmentation	Chapter 4 – Customer Segmentation	RFM segmentation provides an established behavioural framework for evaluating customer value based on purchasing behaviour and complements churn prediction by supporting customer prioritisation.
Decision support strategy	Rule-based prioritisation only	Hybrid decision support integrating business rules with DiCE counterfactual recommendations	Chapter 4 – Decision Support Framework	Business rules determine customer priority using churn probability and RFM segment, while DiCE generates customer-specific recommendations, producing both strategic prioritisation and actionable retention guidance.
Model persistence	Retraining at runtime	Joblib serialisation	Chapter 4 – System Architecture	Serialising trained models reduced computational overhead, ensured consistency between development and deployment environments, and improved dashboard responsiveness.
Deployment platform	Local execution only	Streamlit Community Cloud	Chapter 4 – Deployment	Cloud deployment improved accessibility for demonstration and evaluation without requiring local software installation, enabling stakeholders to interact with the developed artefact through a web-based interface.

Note. The implementation decisions summarised in Table B.1 were guided by the research objectives, the Design Science Research methodology, and the practical requirements of developing an explainable Business Intelligence decision support system for customer retention. Collectively, these decisions balance predictive performance, transparency, usability, and implementation feasibility within a modular decision support architecture. The table is intended to complement the implementation details presented in the main body of the dissertation rather than replace them.

Table B.1 demonstrates that the developed research artefact was constructed through a series of deliberate methodological and technical decisions rather than the arbitrary selection of tools and algorithms. Each implementation choice was informed by the research objectives, supported by evidence presented throughout the dissertation, and aligned with the overarching goal of developing a transparent, explainable, and practically applicable Business Intelligence decision support system for customer retention in retail SMEs.

Appendix C: Customer Behaviour Segmentation and Retention Priority Framework

This appendix documents the customer behaviour segmentation and retention-priority logic implemented within the developed Business Intelligence decision support artefact. The segmentation component applies an adapted Recency–Frequency–Monetary (RFM) framework to organise customers into business-oriented behavioural groups. These segments are subsequently combined with the churn probabilities generated by the Random Forest model to determine whether a customer should receive immediate attention, follow-up, or continued monitoring.

The framework should be interpreted as an adapted RFM model because the source dataset did not contain customer-level transaction revenue, total expenditure, or purchase value. Consequently, CashbackAmount was used as an indirect proxy for the Monetary dimension. The segmentation therefore provides relative behavioural and commercial context within the available dataset rather than a direct measurement of customer revenue contribution or customer lifetime value.

Table C.1
Operationalisation of the Adapted RFM Dimensions

RFM Dimension	Operationalisation in the Artefact	Business Interpretation	Interpretation Limitation
Recency (R)	Represents how recently the customer demonstrated purchasing activity using the recency-related information available in the processed dataset.	More recent activity suggests stronger current engagement, whereas a longer period without activity may indicate weakening customer interest.	Recency reflects the available dataset structure and should be interpreted within the time context represented by the source data.
Frequency (F)	Represents how frequently the customer purchased or interacted using the behavioural information available in the dataset.	Higher frequency indicates more sustained purchasing activity and a potentially stronger customer relationship.	Frequency does not independently indicate profitability because frequent customers may still generate relatively low commercial value.
Monetary proxy (M)	CashbackAmount was used as a proxy because actual transaction value, total customer expenditure, and revenue contribution were unavailable.	Higher CashbackAmount values provide an indirect indication of customer commercial activity and support relative comparison between customers.	CashbackAmount is not equivalent to expenditure, revenue, profitability, or customer lifetime value. Monetary-related segment labels therefore represent relative classifications within the adapted framework.

Note. The implemented segmentation extends conventional RFM logic to accommodate the structure of the available dataset. The Monetary dimension must therefore be interpreted as a proxy-based measure rather than a direct representation of customer spending. This adaptation enabled behavioural segmentation to be incorporated into the artefact while preserving transparency regarding the limitations of the underlying data.

Interpretive clarification

The adapted RFM framework was not used as an alternative churn model. Instead, it provided behavioural context alongside the Random Forest prediction. Churn probability estimated the likelihood of a customer leaving, whereas the behavioural segment indicated the customer's recent activity pattern, purchasing frequency, and relative monetary-proxy position. Combining these outputs enabled the artefact to distinguish between customers who may have similar churn probabilities but different retention significance or engagement histories.

Table C.2

Customer Behaviour Segments Implemented in the Artefact

Customer Behaviour Segment	Behavioural Meaning	Suggested Business Action
High-Value Active Customers	Customers displaying recent activity, frequent purchasing behaviour, and a comparatively high monetary-proxy position.	Reward and retain through loyalty recognition, personalised benefits, or VIP-style engagement.
Frequent Low-Value Customers	Customers who purchase frequently but are classified at a comparatively lower monetary-proxy level.	Increase basket value through relevant bundles, cross-selling, or carefully targeted offers.
High-Value Occasional Customers	Customers classified at a higher monetary-proxy level who purchase less frequently.	Encourage more frequent engagement through personalised communications and relevant incentives.
New and Developing Customers	Recently active customers whose frequency and relative monetary contribution are still developing.	Support early relationship development through onboarding, welcome incentives, and timely communication.
Previously Valuable Customers	Customers who previously demonstrated stronger behavioural or commercial indicators but have become less active.	Use personalised win-back activity to recover the relationship before disengagement becomes permanent.
Frequent Customers Losing Interest	Customers with previously strong purchase frequency whose recent activity has declined.	Re-engage through loyalty incentives, reminder communications, or targeted retention offers.
High-Value Customers Losing Interest	Customers with a comparatively high monetary-proxy position whose recent activity is declining.	Prioritise personalised retention activity and direct engagement because of their relative importance within the segmentation framework.
Inactive Customers	Customers showing weak recent activity, low frequency, and a relatively low monetary-proxy position.	Apply low-cost automated reactivation strategies or continue monitoring where intensive intervention is not commercially justified.

Note. The segment labels are business-facing interpretations of the implemented RFM scoring logic. Terms such as “high-value” refer to relative classifications produced using CashbackAmount as a monetary proxy and should not be interpreted as confirmed revenue, profitability, or customer lifetime value.

Table C.3

Implemented Retention Priority Classification Logic

Churn Probability	Segment Condition	Retention Priority	Decision Interpretation
≥ 0.70	Any behavioural segment	Immediate Action	The predicted churn risk is sufficiently high to justify prompt managerial review regardless of behavioural segment.

0.40–0.69	Previously Valuable Customers	Immediate Action	Moderate churn risk is escalated because the customer previously demonstrated stronger behavioural value but has become less active.
0.40–0.69	Frequent Customers Losing Interest	Immediate Action	Moderate churn risk is escalated because declining recent engagement follows a history of frequent purchasing.
0.40–0.69	High-Value Customers Losing Interest	Immediate Action	Moderate churn risk is escalated because declining activity affects a customer classified at a comparatively high monetary-proxy level.
0.40–0.69	Inactive Customers	Immediate Action	Moderate churn risk combined with inactive behaviour indicates an established pattern of disengagement requiring review.
0.40–0.69	Any remaining behavioural segment	Follow Up Soon	The customer requires timely attention, but the combination of churn risk and segment does not meet the implemented immediate-action conditions.
< 0.40	Any behavioural segment	Monitor	Current predicted churn risk is comparatively low; continued observation is recommended instead of immediate intervention.

Implemented priority logic

The following pseudocode reflects the priority classification implemented in the dashboard:

```

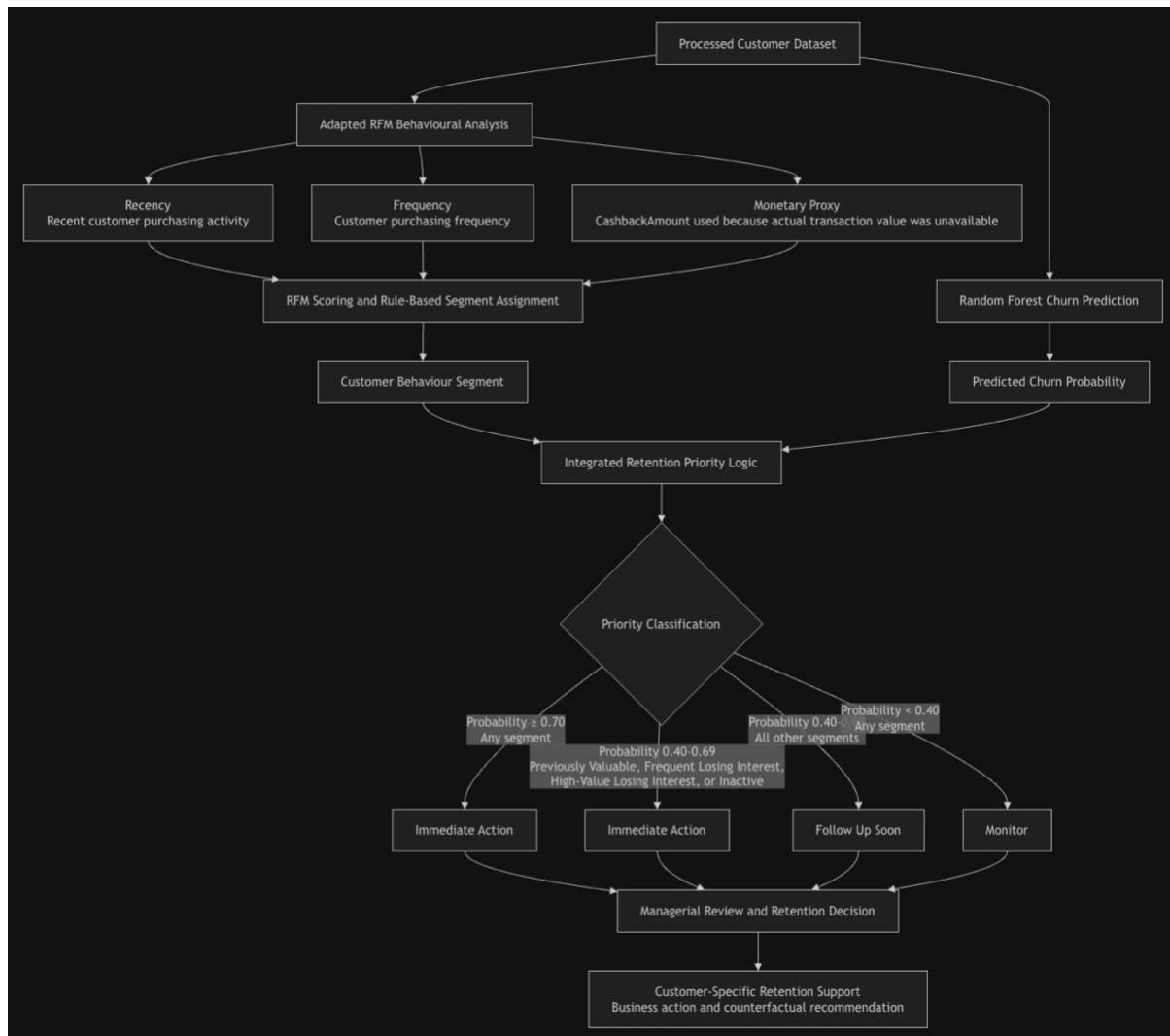
IF churn probability ≥ 0.70
    Priority = Immediate Action

ELSE IF churn probability ≥ 0.40
    IF customer segment is:
        Previously Valuable Customers
        OR Frequent Customers Losing Interest
        OR High-Value Customers Losing Interest
        OR Inactive Customers
    THEN
        Priority = Immediate Action
    ELSE
        Priority = Follow Up Soon

ELSE
    Priority = Monitor

```

This logic demonstrates that the artefact does not determine priority from churn probability alone. For customers within the moderate-risk range, behavioural evidence is used to escalate customers whose segment indicates declining engagement, previous value, or inactivity.

Figure C.1*Integration of Adapted RFM Segmentation and Churn Risk in Retention Prioritisation*

Note. The Monetary component of the adapted RFM framework was operationalised using CashbackAmount as a proxy because the dataset did not contain customer-level transaction revenue or total purchase value. The resulting segments therefore represent relative behavioural and monetary-proxy classifications within the available data. The priority classifications shown in the figure reproduce the implemented dashboard logic: high-risk customers receive immediate attention, while moderate-risk customers are escalated when their behavioural segment indicates previous value, declining interest, or inactivity.

Table C.4*Contribution of Each Analytical Component to the Decision-Support Workflow*

Analytical Component	Output	Decision-Support Contribution
Adapted RFM analysis	Recency, frequency, and monetary-proxy scores	Summarises customer engagement and relative commercial behaviour.
Behavioural segmentation	One of eight customer behaviour segments	Translates numerical behavioural measures into business-readable customer categories.
Random Forest model	Individual churn probability	Estimates the likelihood that a customer may churn.

Priority classification logic	Immediate Action, Follow Up Soon, or Monitor	Converts churn probability and behavioural context into an operational order of attention.
SHAP explanation	Customer-level churn drivers	Explains which variables increased or reduced the individual customer's predicted churn risk.
DiCE counterfactual explanation	Feasible feature-change scenarios	Identifies changes associated with a different predicted outcome and supports customer-specific intervention planning.
Streamlit interface	Interactive profile and decision-support views	Presents analytical outputs in a format intended to support managerial interpretation and action.

Note. The counterfactual component is distinct from the rule-based priority logic. Business rules classify the urgency of managerial attention, whereas DiCE generates customer-specific counterfactual scenarios. Together, these components form a hybrid decision-support approach combining prioritisation rules with explainable, individualised action support.

The customer behaviour segmentation framework adds business context to the predictive component of the artefact. The Random Forest model identifies customers with an elevated likelihood of churn, while the adapted RFM framework distinguishes customers according to engagement patterns and relative monetary-proxy characteristics. The implemented priority rules combine these outputs to translate analytical results into three operational categories: Immediate Action, Follow Up Soon, and Monitor.

This integration strengthens the practical relevance of the artefact because customers with similar churn probabilities may require different levels or forms of intervention depending on their behavioural history. Nevertheless, the use of CashbackAmount as a proxy limits the interpretation of the Monetary dimension. Future implementation using transaction-level sales data would enable conventional monetary-value calculation and provide a stronger basis for estimating customer value, profitability, and retention return on investment.

Appendix D: Explainable AI Decision-Support Workflow

This appendix documents how predictive modelling, SHAP explanations, DiCE counterfactual analysis, and the Streamlit interface operate together within the developed decision-support artefact. The purpose is not to repeat the implementation and evaluation evidence already presented in the main report, but to clarify the distinct contribution of each analytical component and the sequence through which model outputs are translated into business-readable information.

The workflow separates four related questions: whether a customer is likely to churn, which factors contributed to that prediction, what changes are associated with a more favourable predicted outcome, and how a manager should interpret the resulting evidence. This separation is important because predictive probability alone does not provide sufficient guidance for transparent and defensible retention decision-making.

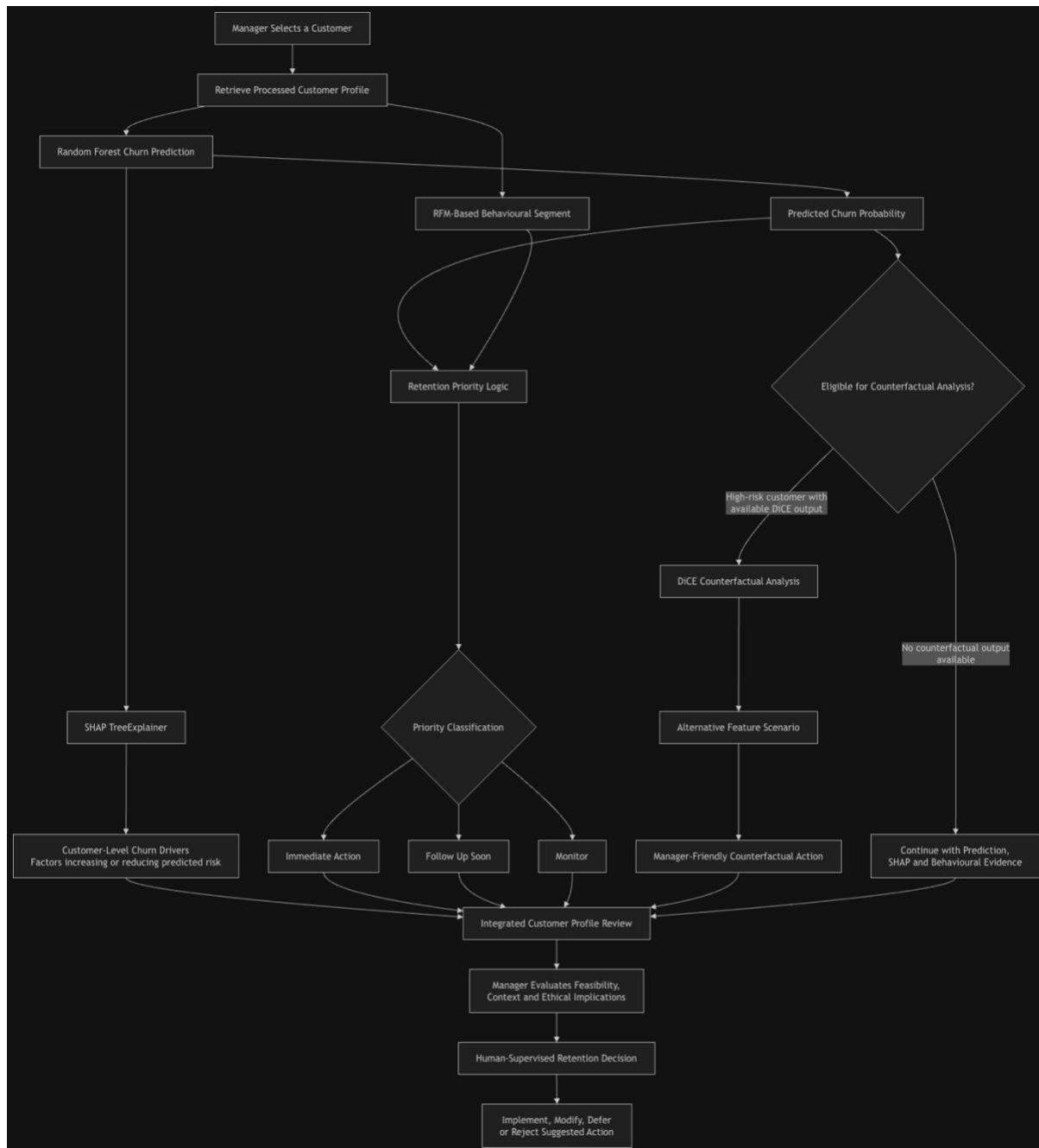
Table D.1

Functions of the Explainable AI Decision-Support Components

Component	Primary Function	Output Presented to the User	Decision-Support Contribution
Random Forest classifier	Estimates the probability that an individual customer will churn.	Customer-level churn probability and risk classification.	Identifies customers who may require retention attention.
RFM-based behavioural segmentation	Provides contextual information about customer engagement and relative monetary-proxy characteristics.	Customer behaviour segment.	Helps managers interpret churn risk in relation to customer behaviour and relative business significance.
SHAP TreeExplainer	Attributes the model prediction to the customer's feature values.	Customer-level factors increasing or decreasing predicted churn risk.	Supports transparency by explaining why the model produced a particular prediction.
DiCE counterfactual analysis	Identifies alternative feature conditions associated with a lower-risk model outcome.	Counterfactual feature changes and manager-readable actions.	Extends explanation towards intervention planning by indicating what could potentially change.
Priority classification logic	Combines churn probability and behavioural segment using implemented business rules.	Immediate Action, Follow Up Soon, or Monitor.	Converts analytical outputs into an operational order of managerial attention.
Streamlit dashboard	Integrates and communicates the analytical outputs through interactive views.	Executive summaries, customer profiles, churn drivers, priorities, and counterfactual actions.	Enables non-technical users to access and interpret model-supported retention evidence.
Human managerial judgement	Reviews analytical evidence in relation to customer circumstances and organisational constraints.	Final retention decision.	Preserves human oversight and prevents model outputs from being treated as automatic decisions.

Note. The Random Forest, SHAP, and DiCE components perform different functions. SHAP explains the existing prediction, whereas DiCE identifies alternative scenarios associated with a different predicted outcome. Neither technique independently determines whether a retention intervention should be implemented; the final decision remains subject to managerial review.

Figure D.1
Integrated Explainable AI Decision-Support Workflow



Note. The workflow distinguishes prediction, explanation, prioritisation, and counterfactual action support. SHAP identifies the variables contributing to the current churn prediction, while DiCE produces alternative model scenarios for customers with available counterfactual outputs. The managerial review stage is retained because counterfactual results indicate model-associated possibilities rather than guaranteed causal effects or automatically executable interventions.

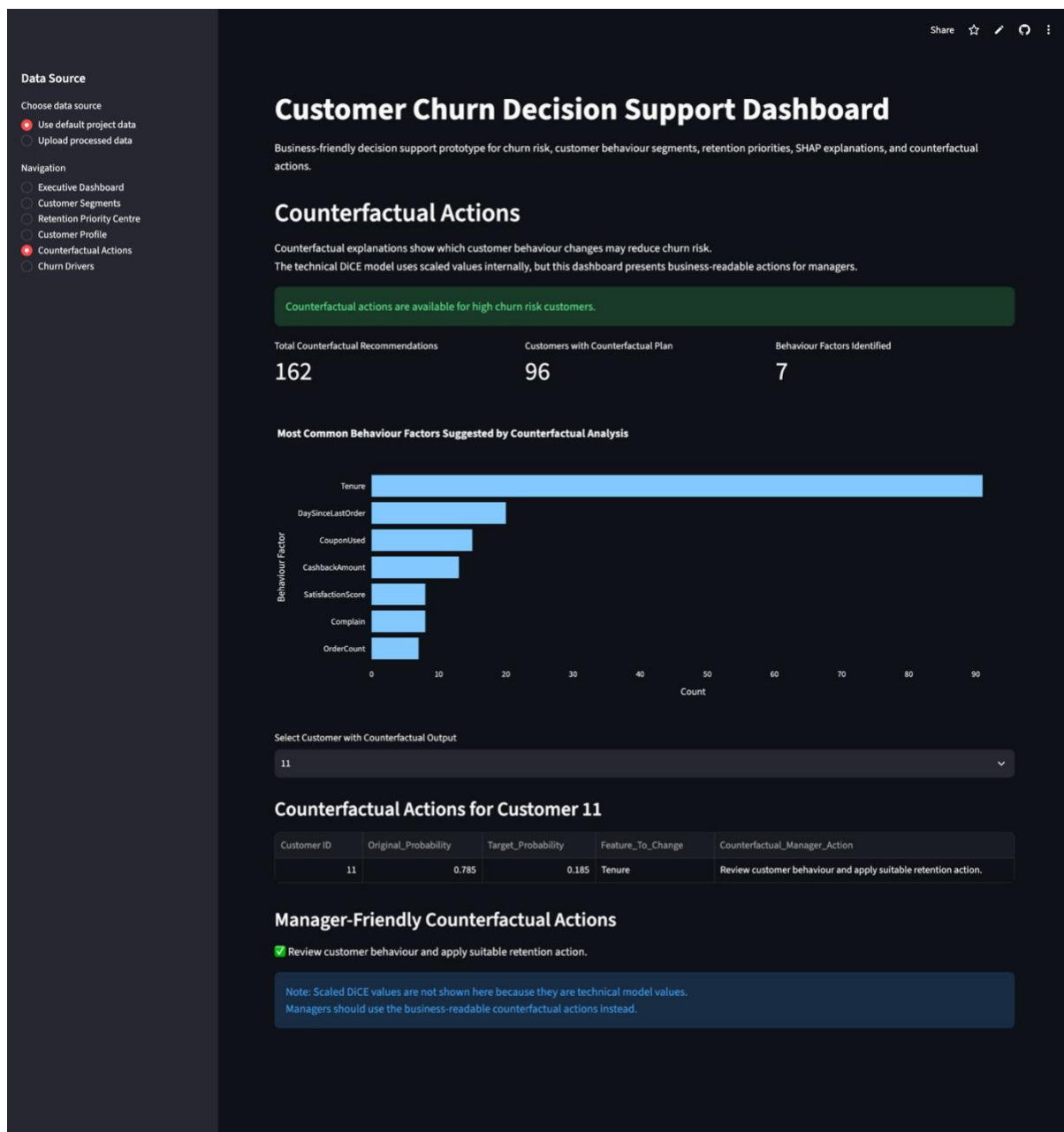
Table D.2
Managerial Interpretation Guide for Dashboard Outputs

Dashboard Output	Appropriate Managerial Interpretation	Interpretation Caution
High predicted churn probability	The model estimates that the customer has a comparatively high likelihood of belonging to the churn class.	The value is a model-based estimate, not certainty that the customer will leave.
Feature increasing predicted risk in SHAP output	The customer's value for that feature contributed towards a higher churn prediction relative to the model baseline.	The feature contribution should not automatically be interpreted as causal.
Feature reducing predicted risk in SHAP output	The feature acted as a protective influence within the model prediction.	A negative contribution does not guarantee customer retention.
Global SHAP feature importance	The feature is influential across the analysed customer population.	Global importance does not explain every individual customer in the same way.
Immediate Action	The implemented priority rules indicate that the customer requires prompt managerial review.	It does not mandate a particular intervention or expenditure.
Follow Up Soon	The customer requires timely attention, although the implemented conditions do not classify the case as the highest priority.	Managers should still consider customer value, feasibility, and available resources.
Monitor	The current model output and priority logic do not indicate a need for immediate action.	Monitoring should not be interpreted as permanently excluding the customer from future intervention.
Counterfactual target probability	The counterfactual scenario is associated with a lower predicted churn probability under the model.	It is a hypothetical model output rather than proof that implementing the change will cause the target probability.
Feature suggested for change	DiCE identified the feature as part of an alternative scenario associated with a different classification or probability.	Some feature changes may be immutable, impractical, costly, ethically inappropriate, or outside managerial control.
Manager-friendly action	A business-readable interpretation of the counterfactual output intended to support managerial consideration.	It should be adapted to the customer context rather than applied mechanically.

Note. The interpretation guide is intended to reduce the risk that managers treat predictive and explanatory outputs as deterministic instructions. Appropriate use requires professional judgement, consideration of feasibility, and awareness of the limitations of model-derived evidence.

Figure D.2

Counterfactual Actions Page Presenting Aggregate Patterns and Customer-Level Action Support



Note. The Counterfactual Actions page presents the availability and distribution of counterfactual outputs, identifies the behaviour factors most frequently involved in generated scenarios, and enables the user to inspect a selected customer's original probability, target probability, feature identified for change, and manager-readable action. The displayed example for Customer 11 shows an original predicted probability of 0.785 and a counterfactual target probability of 0.185, with Tenure identified as the feature associated with the alternative scenario. The interface deliberately suppresses scaled DiCE values because they are technical modelling outputs and instead presents business-readable guidance.

Table D.3
Interpretation of the Counterfactual Actions Page

Interface Element	Evidence Displayed in Figure D.2	Decision-Support Purpose
Counterfactual availability message	Counterfactual actions are available for high churn-risk customers	Communicates the scope of the counterfactual module and prevents users from expecting outputs for every customer.
Total counterfactual recommendations	162 recommendations	Indicates the overall volume of generated action scenarios available in the deployed artefact.
Customers with a counterfactual plan	96 customers.	Shows how many customers received at least one generated counterfactual output.
Behaviour factors identified	Seven factors.	Summarises the range of variables involved across generated counterfactual scenarios.
Aggregate factor chart	Tenure appears most frequently, followed by DaySinceLastOrder, CouponUsed, CashbackAmount, SatisfactionScore, Complain, and OrderCount	Provides a portfolio-level view of which factors recur most often within the generated alternative scenarios.
Customer selector	Customer 11 is selected in the example.	Supports customer-level inspection rather than relying only on aggregated results.
Original probability	0.785	Shows the model's predicted churn risk before the alternative scenario.
Target probability	0.185	Shows the predicted probability associated with the generated counterfactual scenario.
Feature to change	Tenure	Identifies the feature involved in the alternative model scenario.
Manager action	Review customer behaviour and apply a suitable retention action	Converts technical output into a cautious, business-readable prompt for managerial review.
Technical-value notice	Scaled DiCE values are not shown	Prevents non-technical users from mistaking transformed model values for directly meaningful business measures.

Note. Aggregate counterfactual frequencies show how often a feature appeared within generated scenarios; they do not demonstrate that the feature is the most effective intervention or that changing it will causally reduce churn. The example probabilities describe the behaviour of the trained model under alternative feature conditions.

Table D.4
Questions Addressed by the Integrated Analytical Components

Analytical Component	Principal Question Addressed	What It Does Not Establish
Random Forest prediction	How likely is this customer to churn according to the trained model?	It does not explain the prediction or prescribe an intervention.
RFM behavioural segment	What type of behavioural relationship does the customer exhibit?	It does not independently predict churn or determine causal customer value.
SHAP explanation	Why did the model assign this level of churn risk?	It does not demonstrate that the identified factors cause churn.
Priority logic	How urgently should the customer be reviewed under the implemented rules?	It does not determine the final business action.

DiCE counterfactual	What alternative feature conditions are associated with a more favourable model outcome?	It does not guarantee that the changes are feasible, ethical, causal, or commercially effective.
Dashboard integration	How can these outputs be presented within one managerial workflow?	It does not replace organisational policies, customer knowledge, or professional judgement.
Managerial oversight	What action is appropriate after considering the available analytical and contextual evidence?	It cannot eliminate uncertainty or limitations in the underlying data and model.

Table D.5

Human-Oversight Checks Before Applying Counterfactual Guidance

Review Area	Managerial Check
Actionability	Can the organisation realistically influence the suggested feature?
Mutability	Is the feature capable of changing, or does it represent a fixed or historical characteristic?
Causal plausibility	Is there credible business reasoning that the intervention could influence customer behaviour, rather than merely changing a model input?
Customer appropriateness	Is the action suitable for the individual customer's circumstances and relationship history?
Fairness	Could applying the recommendation create unjustified differences in treatment between customer groups?
Privacy	Does the proposed use of customer information comply with applicable consent, access, and governance requirements?
Commercial feasibility	Is the likely retention value proportionate to the cost and operational effort of the intervention?
Consistency with policy	Does the action comply with organisational retention, marketing, and customer-service policies?
Monitoring	Can the organisation observe the outcome and determine whether the intervention was effective?

Note. These checks position the artefact as a human-centred decision-support system rather than an automated decision-maker. They also align the use of explainability with responsible AI principles of transparency, accountability, fairness, privacy, and human oversight.

The explainable AI workflow demonstrates that the artefact extends beyond churn prediction by integrating behavioural context, feature-level explanation, priority classification, and counterfactual action support within a single managerial interface. Random Forest estimates churn risk, SHAP explains the factors associated with the prediction, and DiCE identifies alternative model scenarios that may inform intervention planning. The Streamlit dashboard then presents these outputs in a form intended to support interpretation by non-technical users.

The contribution of this workflow lies in the separation and integration of predictive, explanatory, and action-oriented functions. However, the outputs remain model-dependent and should not be interpreted as causal instructions. Counterfactual actions require managerial assessment of feasibility, fairness, customer appropriateness, cost, and organisational policy before implementation. Human oversight therefore remains a central component of the developed decision-support approach.

Appendix E: User Evaluation Questionnaire

The following Microsoft Forms questionnaire was used to collect anonymous participant feedback after users explored the developed customer churn decision support dashboard. The instrument included participant-profile questions, five-point Likert-scale statements, and open-ended questions concerning usability, explainability, recommendation usefulness, and potential improvements.

Figure E.1

Prototype Evaluation Form for the Customer Churn Decision Support Dashboard

Prototype Evaluation Form: Customer Churn Decision Support Dashboard

Thank you for participating in this evaluation. This survey is part of a Master's research project evaluating an Explainable AI Customer Retention Decision Support Dashboard for retail SMEs. You will first explore the dashboard and then answer a few short questions about its usability, explainability, and usefulness. Your responses are anonymous and will be used only for academic research. Estimated completion time: **3–5 minutes**.

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

* Required

1. What best describes your role? *

- ☐ Business student
- ☐ Small business owner
- ☐ Retail manager
- ☐ Business/analytics professional
- ☐ Other

How familiar are you with business dashboards or analytics tools? *

- ☐ Not familiar
- ☐ Slightly familiar
- ☐ Moderately familiar
- ☐ Very familiar

3. The dashboard was easy to navigate and use. *

1	2	3	4	5
---	---	---	---	---

4. The dashboard helped me identify customers who require retention action. *

1	2	3	4	5
---	---	---	---	---

5. The explanation helped me understand the main reasons why the system predicted that a customer might churn. *

1	2	3	4	5
---	---	---	---	---

6. The counterfactual recommendations were useful for identifying possible retention actions. *

1	2	3	4	5
---	---	---	---	---

7. Which feature of the dashboard did you find most useful, and why? *

Enter your answer

8. Was there anything you found confusing or difficult to understand? *

Enter your answer

9. What improvements would you suggest for future versions of this dashboard? *

Enter your answer

10. Overall, I would consider using this dashboard to support customer retention decisions. *

1	2	3	4	5
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Submit

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Note. Microsoft Forms questionnaire developed by the author for the formative user evaluation. Participants evaluated dashboard navigation, identification of customers requiring retention action, understanding of churn explanations, usefulness of counterfactual recommendations, and willingness to use the dashboard for customer retention decision-making.

Appendix F: Participant Profile

The formative user evaluation involved 11 anonymous participants representing both business-oriented and technically informed perspectives. Participants were asked to identify their professional role and level of familiarity with business dashboards or analytics tools before evaluating the prototype. No personally identifiable information was collected. Tables F.1 and F.2 summarise the participant characteristics.

Table F.1

Participant Roles

Participant Role	Frequency	Percentage
Retail manager	5	45.5%
Business student	4	36.4%
Small business owner	1	9.1%
Data Engineer	1	9.1%
Total	11	100.0%

Note. Participant roles were self-reported through the Microsoft Forms evaluation questionnaire. Percentages are rounded to one decimal place.

Table F.2

Participant Familiarity with Business Dashboards or Analytics Tools

Familiarity Level	Frequency	Percentage
Not familiar	1	9.1%
Slightly familiar	3	27.3%
Moderately familiar	4	36.4%
Very familiar	3	27.3%
Total	11	100.0%

Note. Familiarity levels were self-reported by participants prior to evaluating the developed dashboard.

Appendix G: Anonymous User Evaluation Responses

This appendix presents the anonymous qualitative feedback collected during the formative user evaluation of the developed customer churn decision support dashboard. The responses informed the thematic analysis reported in Chapter 5 and have been reproduced without identifying participants. Minor spelling and grammatical errors have been retained to preserve the authenticity of participant responses.

Table G.1

Participant Responses: Most Useful Dashboard Feature

Participant	Response
P01	Customer profile because it provides a clear overview of each customer and supports appropriate retention actions.
P02	Retention Priority Centre because it clearly identifies customers requiring immediate attention.
P03	Customer segmentation because it helps understand different customer groups.
P04	Understanding how customers can be retained.
P05	Customer segmentation.
P06	Churn drivers.
P07	Recommendation section.
P08	Action recommendations.
P09	Customer grouping, segmentation, and retention priority.
P10	Filtering features.
P11	All dashboard features.

Note. Responses have been lightly edited only for readability where necessary while preserving their original meaning.

Table G.2

Participant Responses: Difficulties or Confusing Aspects

Participant	Response
P01	Segmentation chart was difficult to understand.
P02	Dashboard was generally easy to understand.
P03	Straightforward to use.
P04	No difficulties reported.
P05	No difficulties reported.
P06	Overall layout and technical language could be improved.
P07	No difficulties reported.
P08	No difficulties reported.
P09	Nothing confusing.
P10	No difficulties reported.
P11	Churn analysis could be explained more clearly.

Table G.3

Participant Suggestions for Future Improvements

Participant	Suggestion
P01	Display all customer information on a single page.
P02	Add real-time data updates.
P03	Simplify the dashboard layout.

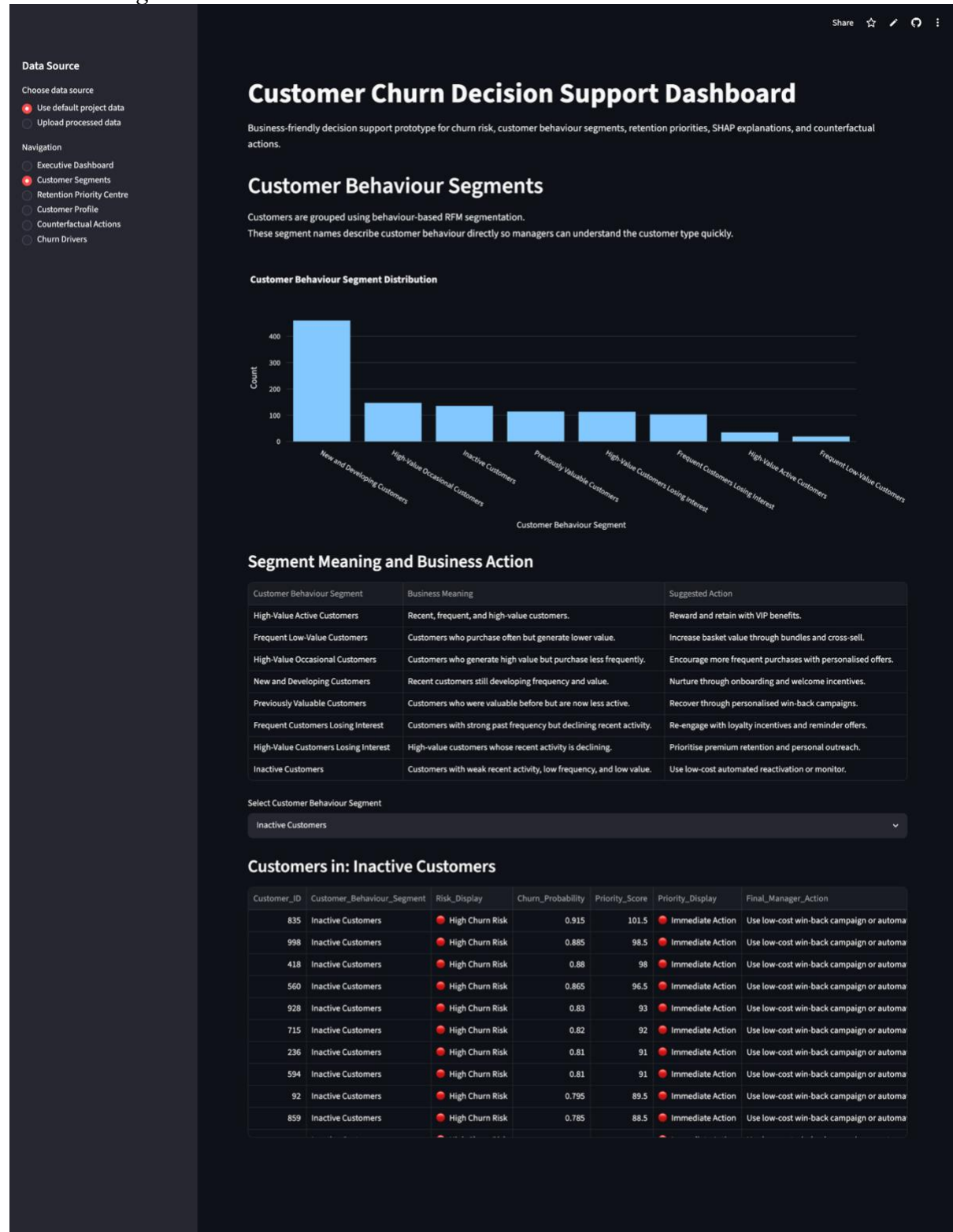
P04	No further improvements suggested.
P05	Improve customer contact information management.
P06	Make the interface more user-friendly.
P07	Improve overall usability.
P08	No suggestions provided.
P09	Enhance customer grouping functionality.
P10	Support additional file upload formats such as PDF and text files.
P11	No additional improvements suggested.

Appendix H: Additional Dashboard Screenshots

This appendix presents supplementary screenshots of the developed Streamlit-based customer churn decision support dashboard that are not included in the main report. These interfaces provide additional evidence of the implemented functionality described in Chapter 4.

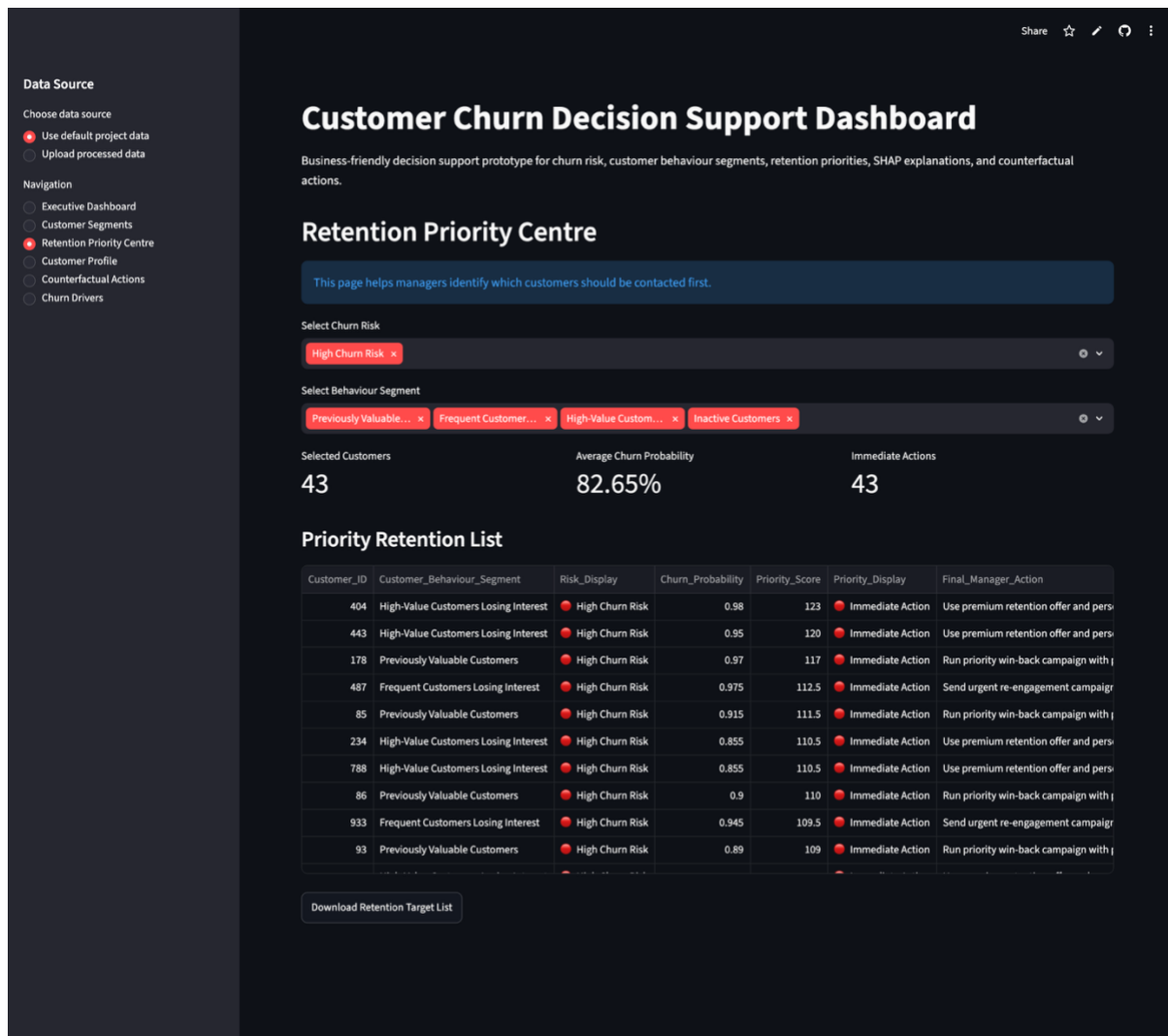
Figure H.1

Customer Segmentation Dashboard



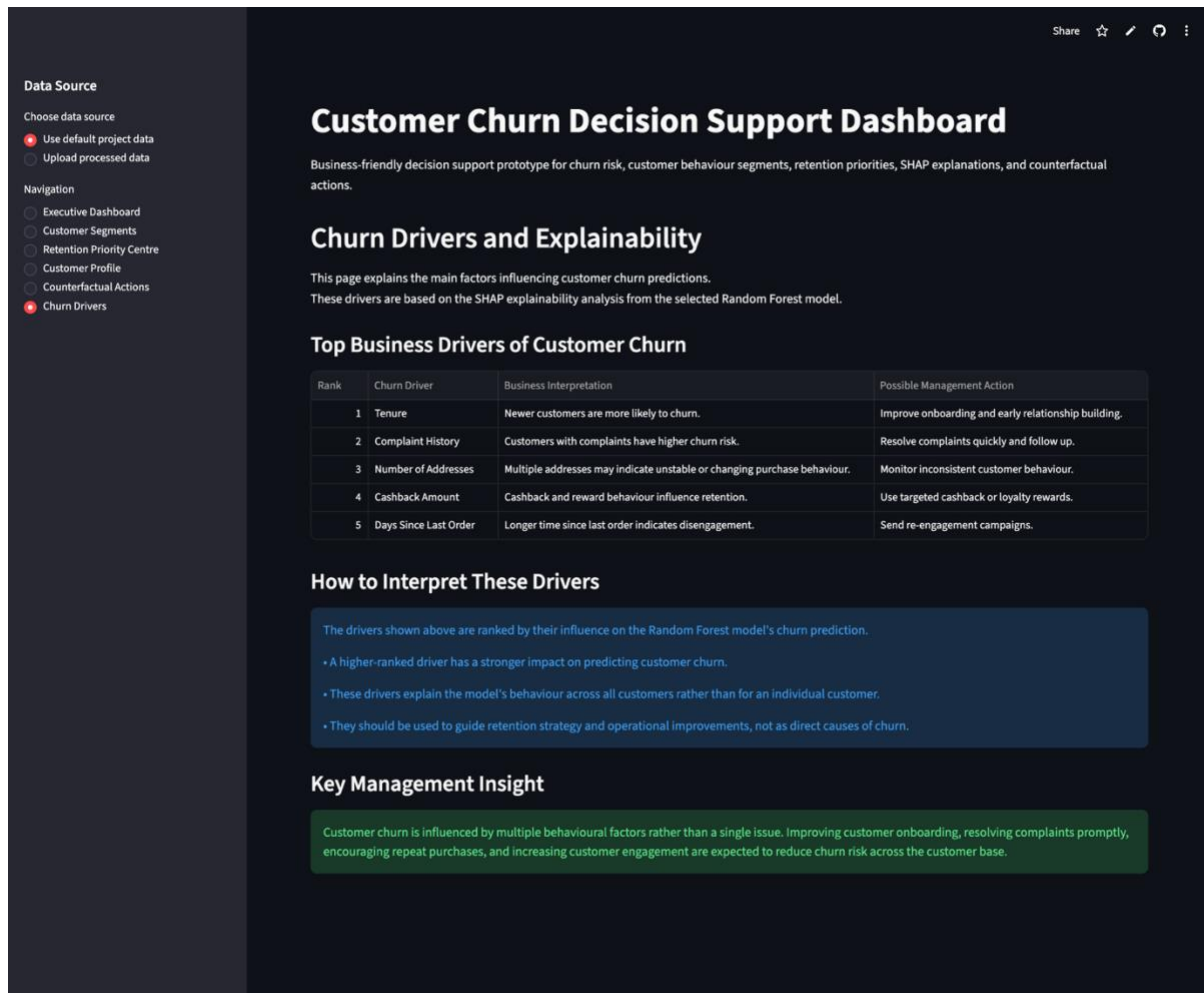
Note. Displays customer behavioural segments generated using the adapted RFM analysis to support customer profiling and retention strategy development.

Figure H.2
Retention Priority Centre



Note. Displays customers prioritised according to predicted churn probability, behavioural segmentation, and the rule-based priority classification (Immediate Action, Follow Up Soon, and Monitor), enabling managers to identify customers requiring timely retention interventions.

Figure H.3
SHAP Explainability Dashboard



Note. Illustrates the global feature importance used to improve the transparency and interpretability of customer churn predictions.

Figure H.4
Processed Data Upload Interface

The screenshot displays the 'Customer Churn Decision Support Dashboard' with a dark theme. On the left sidebar, under 'Data Source', there are two radio buttons: 'Use default project data' and 'Upload processed data' (which is selected). Below this, a blue box contains the text: 'Prototype upload mode: please upload processed CSV files only. Raw Excel upload is not supported in this version.' Further down, there are three upload sections, each with an 'Upload' button and a '200MB per file • CSV' limit. The first section is for 'final_recommendation_engine_output.csv', the second for 'rfm_segmented_data.csv', and the third (labeled 'Optional: Upload') is for 'dice_counterfactual_highrisk.csv'. The main content area has the title 'Customer Churn Decision Support Dashboard' and a subtitle: 'Business-friendly decision support prototype for churn risk, customer behaviour segments, retention priorities, SHAP explanations, and counterfactual actions.' A green box in the main area prompts the user: 'Please upload the recommendation CSV and RFM CSV to continue.' The top right corner of the dashboard includes icons for 'Share', a star, a pencil, a speech bubble, and a menu.

Note. The prototype includes a data upload interface that demonstrates how processed datasets can be supplied to the dashboard. Users are required to generate the recommendation, RFM segmentation, and optional counterfactual output files using the external preprocessing and machine learning pipeline before uploading the corresponding CSV files. As this is a prototype, the interface illustrates the intended data integration workflow rather than providing automated model training or raw data processing within the application.

Appendix I: Dataset Data Dictionary

This appendix summarises the variables contained in the e-commerce customer churn dataset used throughout the implementation and evaluation of the developed decision support system. The dataset comprises customer demographic, behavioural, transactional, and service-related attributes that were used during data preprocessing, feature engineering, customer segmentation, churn prediction, and explainable AI analysis. Variable descriptions are adapted from the original dataset documentation.

Table I.1
Dataset Data Dictionary

Data Source	Variable	Description
E-Commerce Customer Dataset	CustomerID	Unique customer identifier
E-Commerce Customer Dataset	Churn	Customer churn flag (target variable)
E-Commerce Customer Dataset	Tenure	Length of customer relationship with the organisation
E-Commerce Customer Dataset	PreferredLoginDevice	Customer's preferred login device
E-Commerce Customer Dataset	CityTier	Customer's city tier classification
E-Commerce Customer Dataset	WarehouseToHome	Distance between the warehouse and the customer's home
E-Commerce Customer Dataset	PreferredPaymentMode	Customer's preferred payment method
E-Commerce Customer Dataset	Gender	Customer gender
E-Commerce Customer Dataset	HourSpendOnApp	Average hours spent using the mobile application or website
E-Commerce Customer Dataset	NumberOfDeviceRegistered	Number of devices registered by the customer
E-Commerce Customer Dataset	PreferredOrderCat	Customer's preferred product category during the previous month
E-Commerce Customer Dataset	SatisfactionScore	Customer satisfaction score
E-Commerce Customer Dataset	MaritalStatus	Customer marital status
E-Commerce Customer Dataset	NumberOfAddress	Number of addresses associated with the customer account

E-Commerce Customer Dataset	Complain	Indicates whether the customer submitted a complaint during the previous month
E-Commerce Customer Dataset	OrderAmountHikeFromlastYear	Percentage increase in order amount compared with the previous year
E-Commerce Customer Dataset	CouponUsed	Number of coupons used during the previous month
E-Commerce Customer Dataset	OrderCount	Number of orders placed during the previous month
E-Commerce Customer Dataset	DaySinceLastOrder	Number of days since the customer's last order
E-Commerce Customer Dataset	CashbackAmount	Average cashback received during the previous month <i>(used as a proxy for the Monetary component of the adapted RFM segmentation in this research)</i>

Note. The original dataset did not contain a transaction value or purchase amount variable. Consequently, CashbackAmount was used as a proxy for the Monetary component when implementing the adapted RFM customer segmentation framework. This limitation is discussed in Section 6.3.1.